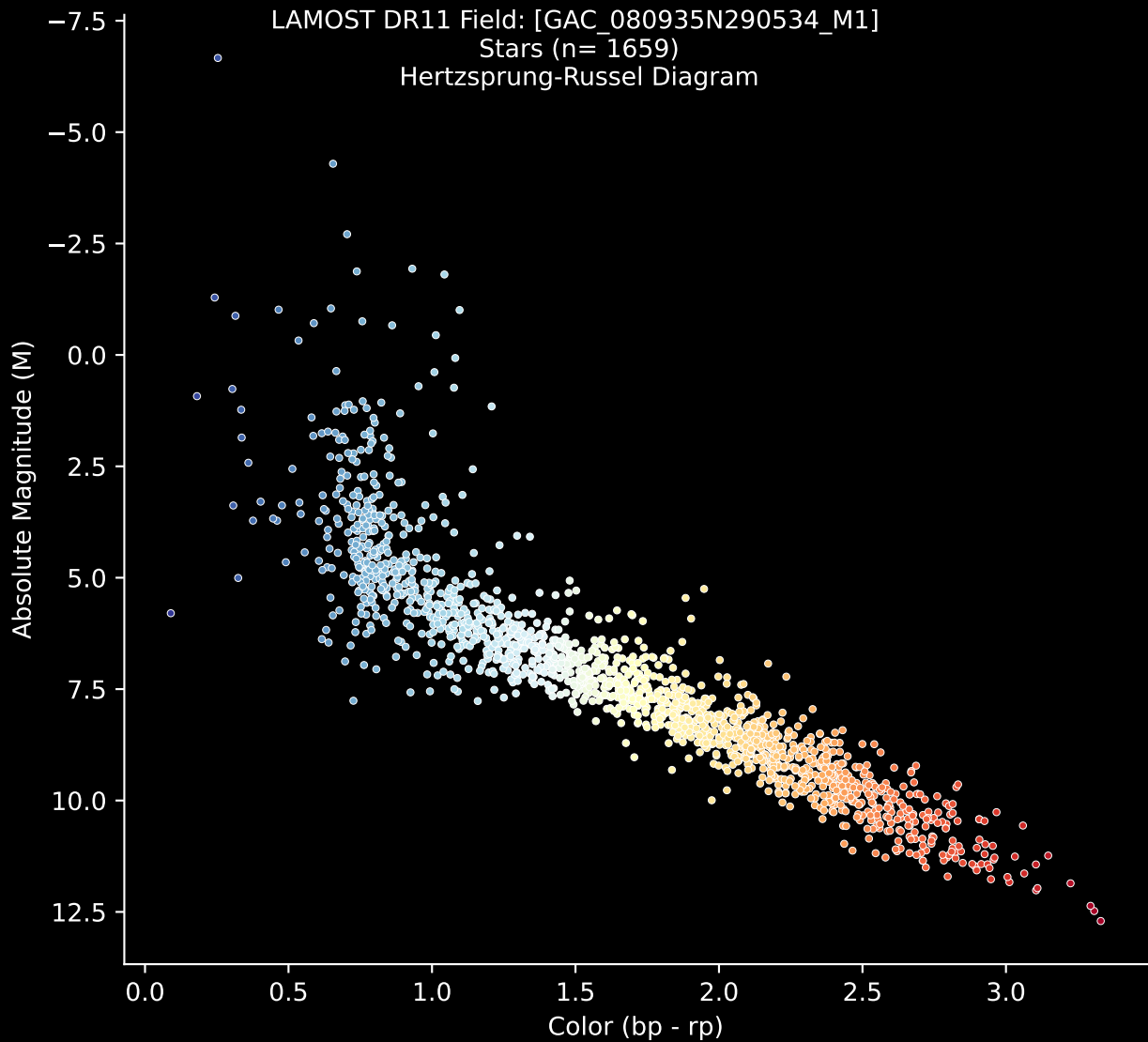
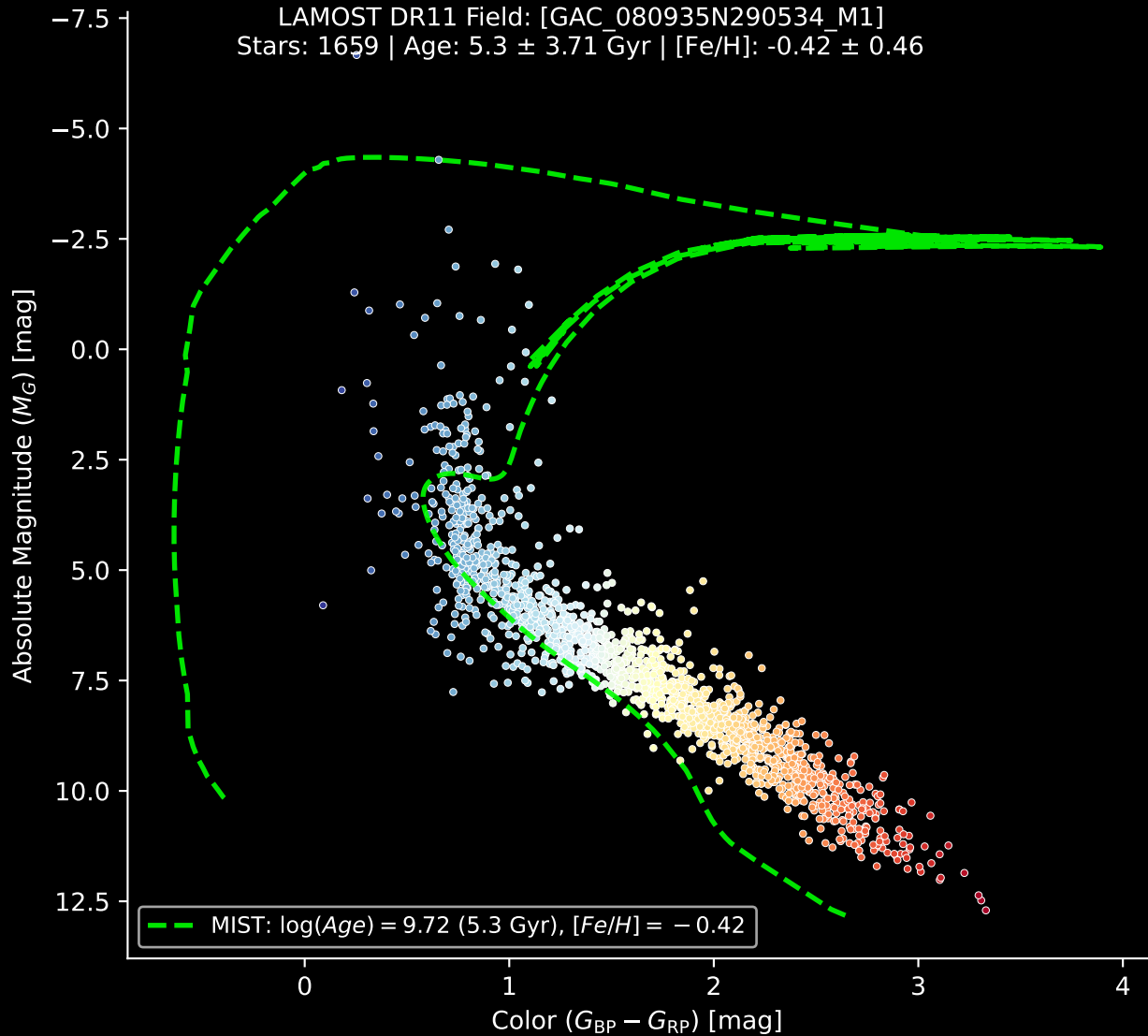


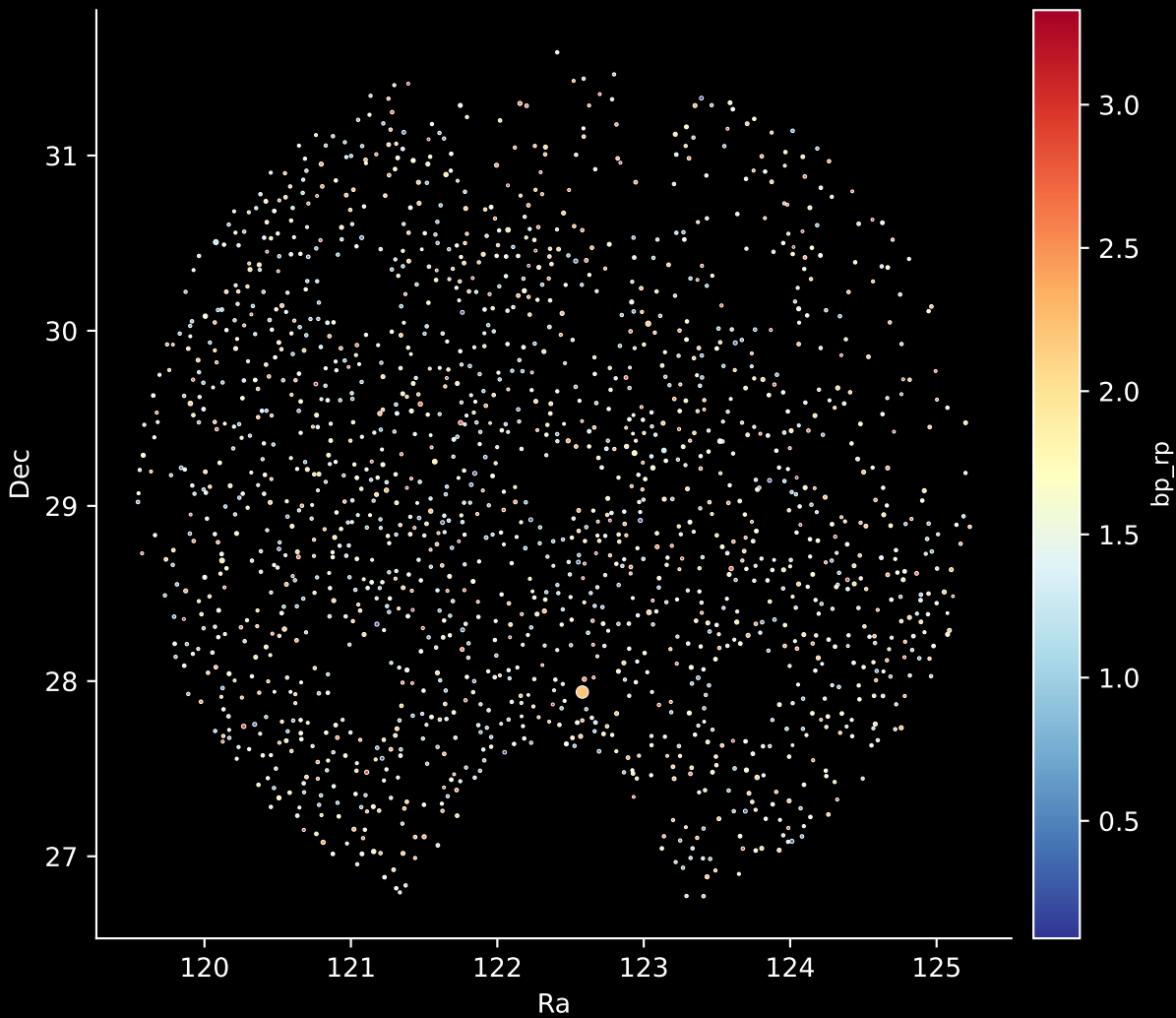


LAMOST DR11 Field: [GAC_080935N290534_M1]
Stars: 1659 | Age: 5.3 ± 3.71 Gyr | $[Fe/H]: -0.42 \pm 0.46$



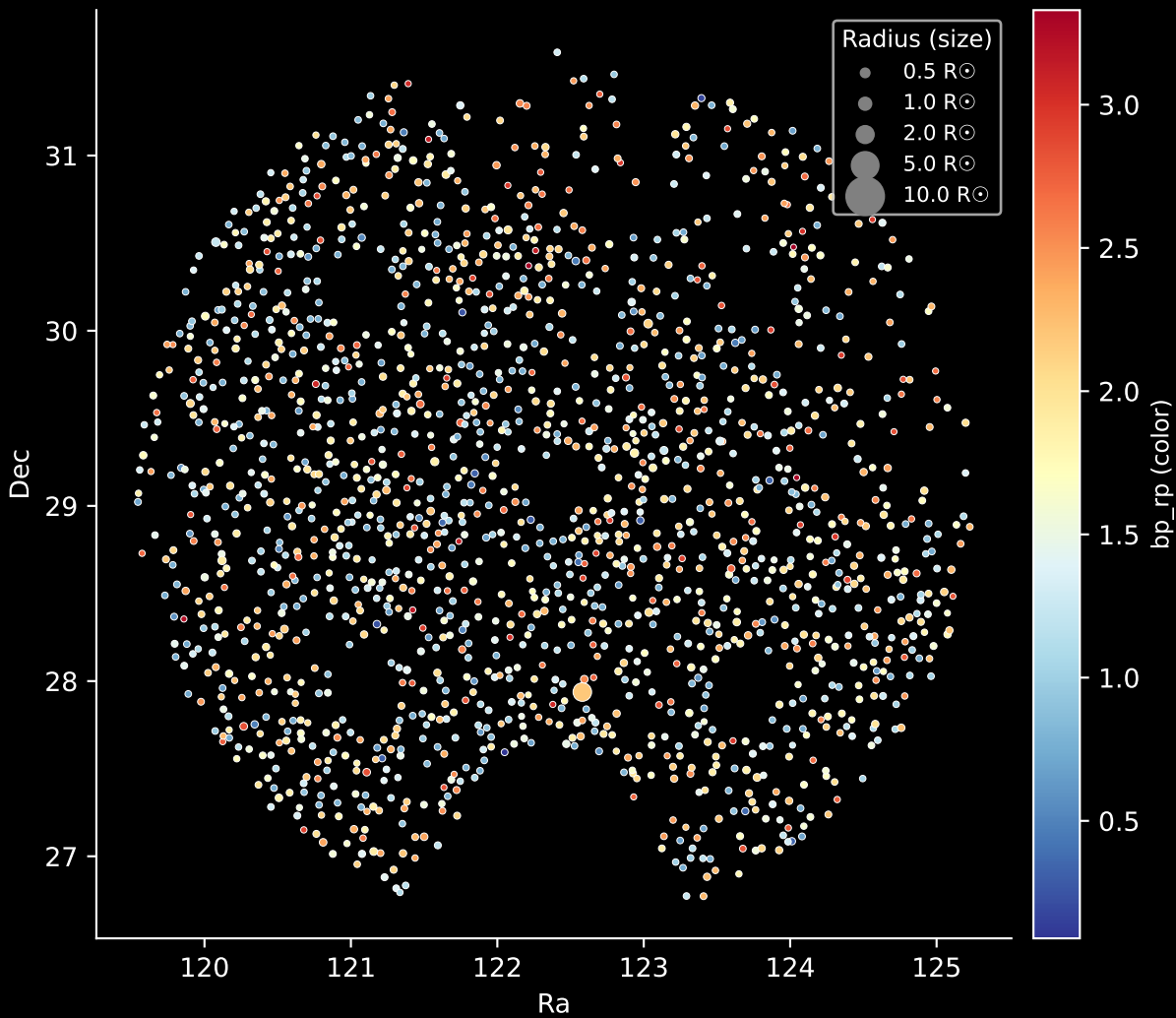
LAMOST DR11 Field: [GAC_080935N290534_M1]

Stars: 1659

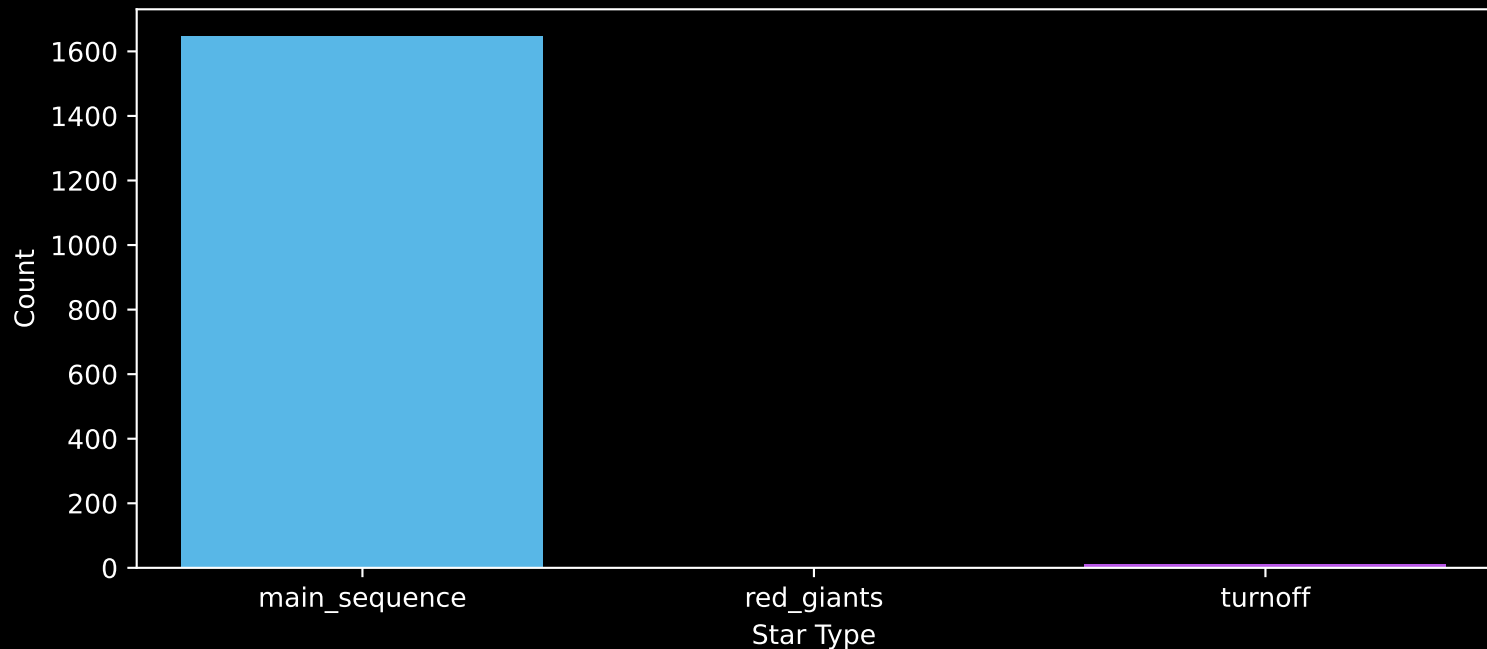


LAMOST DR11 Field: [GAC_080935N290534_M1]

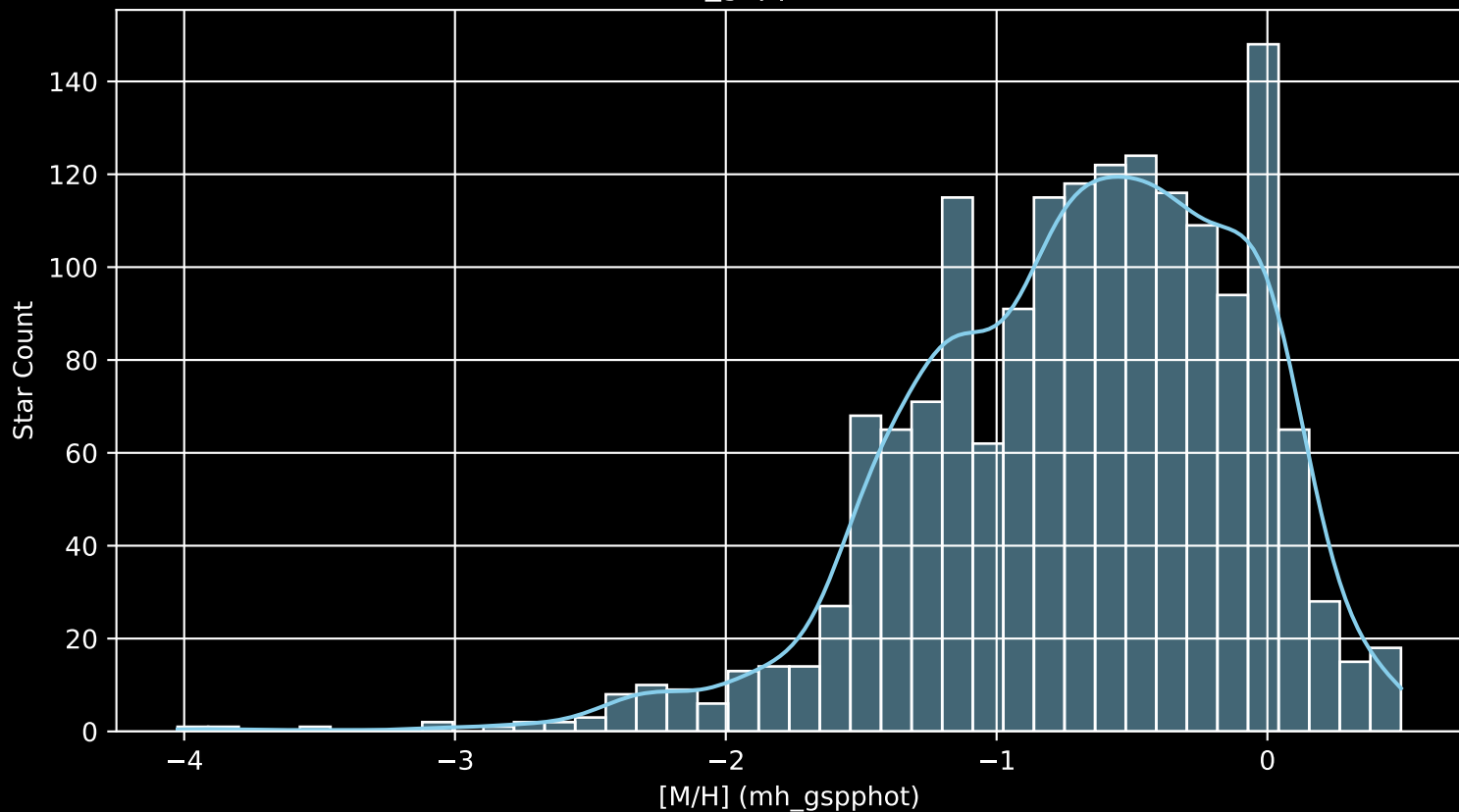
Stars: 1659



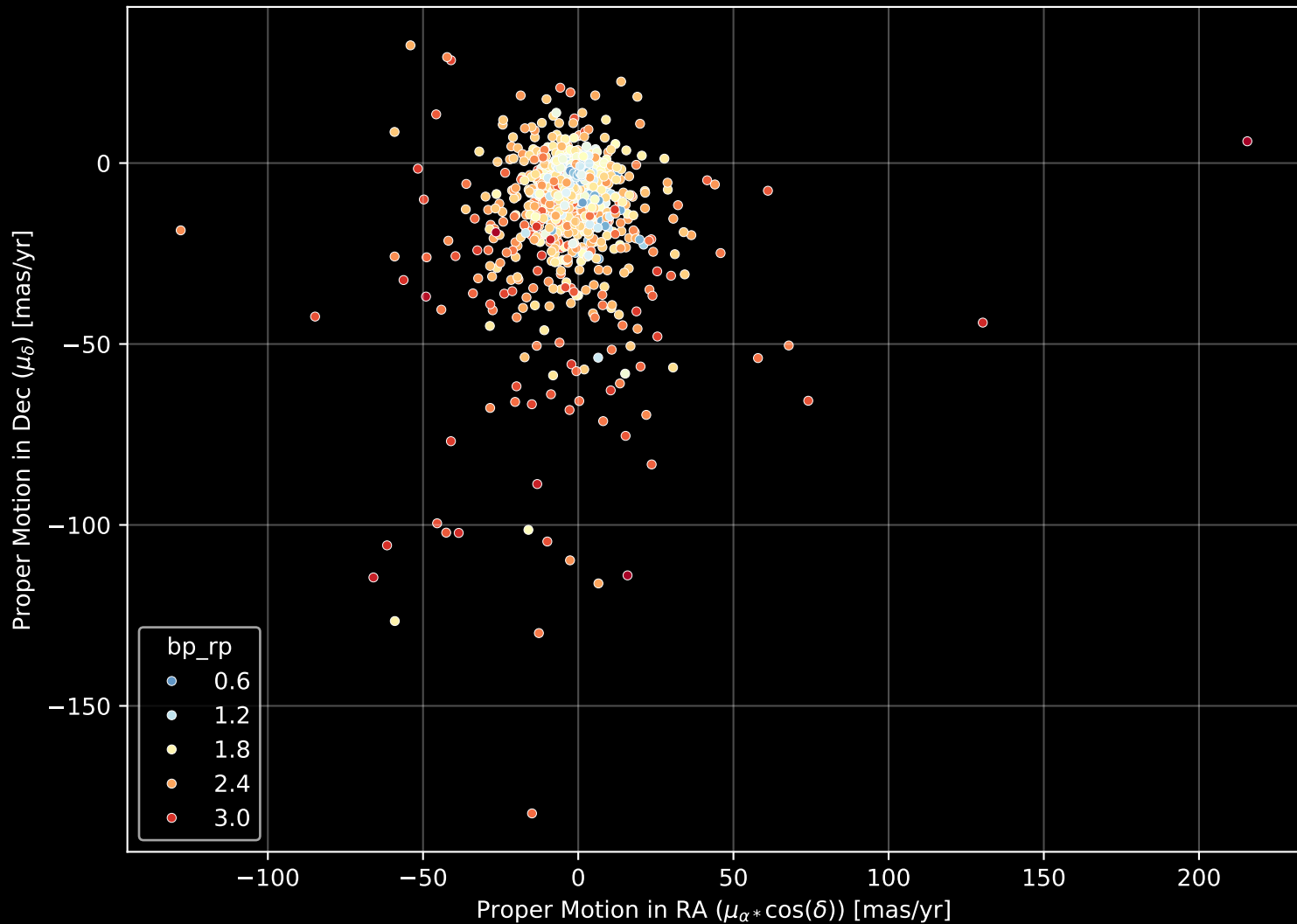
LAMOST DR11 Field: [GAC_080935N290534_M1]
Count plot of Star Type



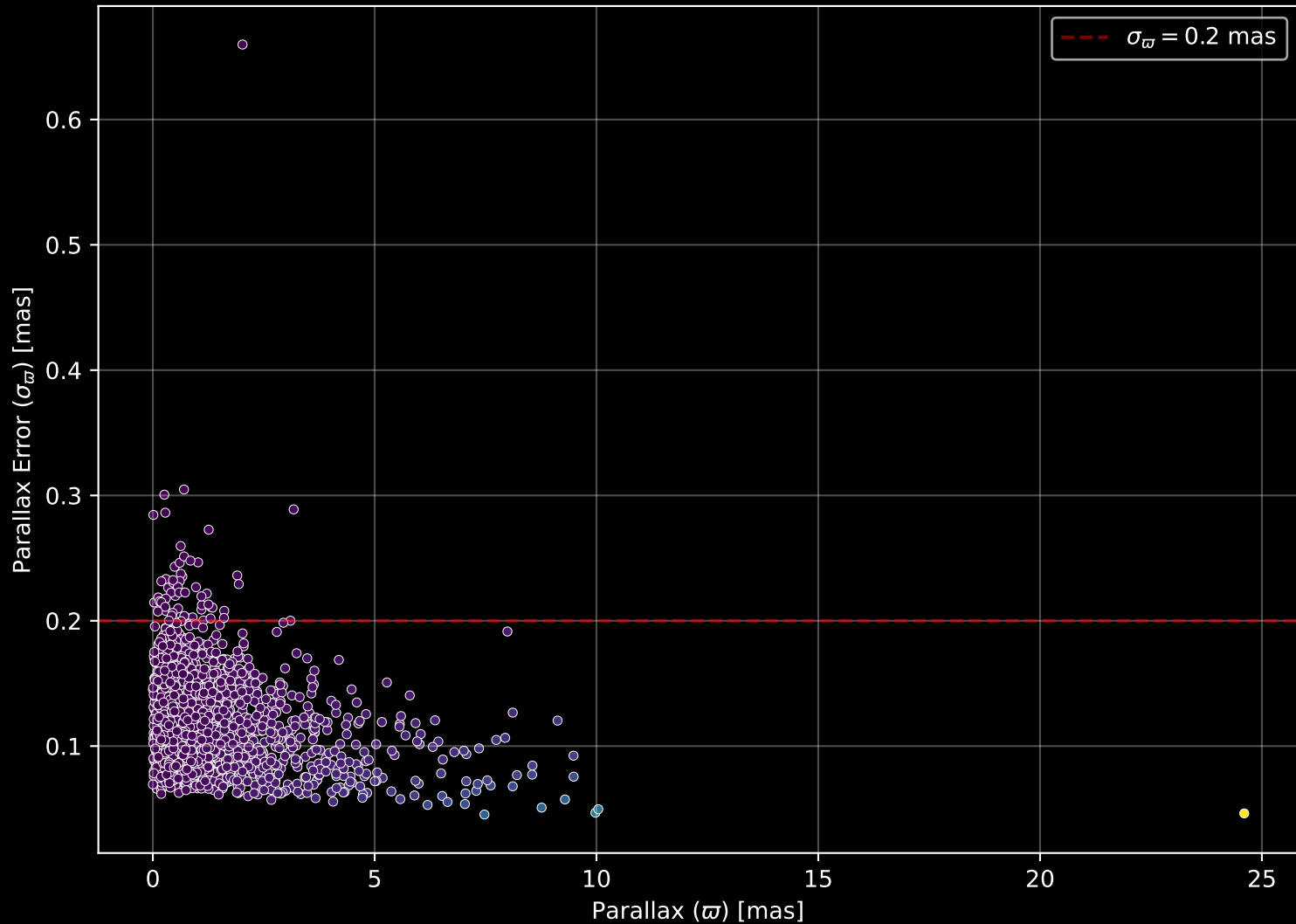
LAMOST DR11 Field: [GAC_080935N290534_M1
[M/H] (mh_gspphot) (n= 1658)



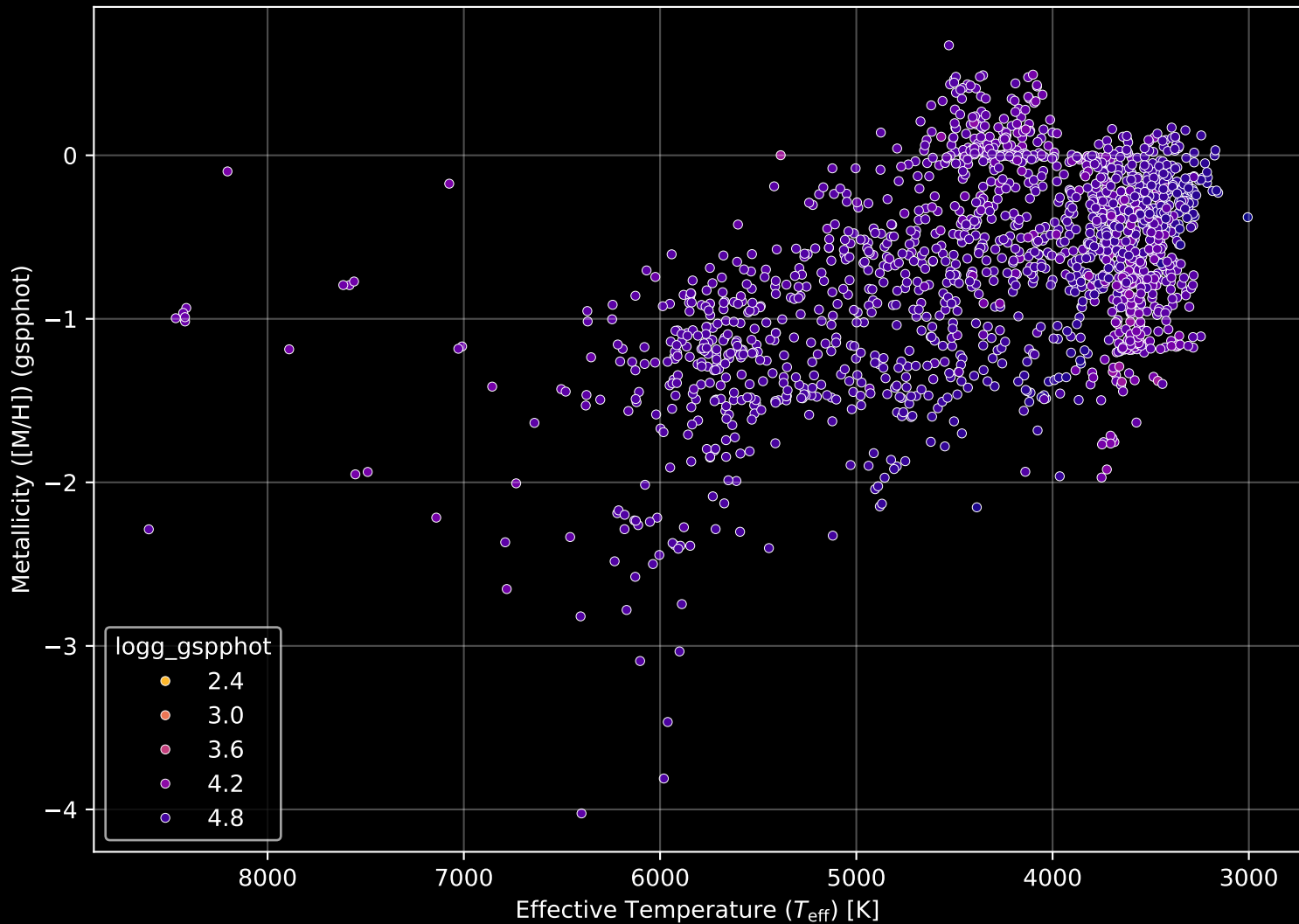
LAMOST DR11 Field: [GAC_080935N290534_M1] Proper Motion Diagram (n= 1659)



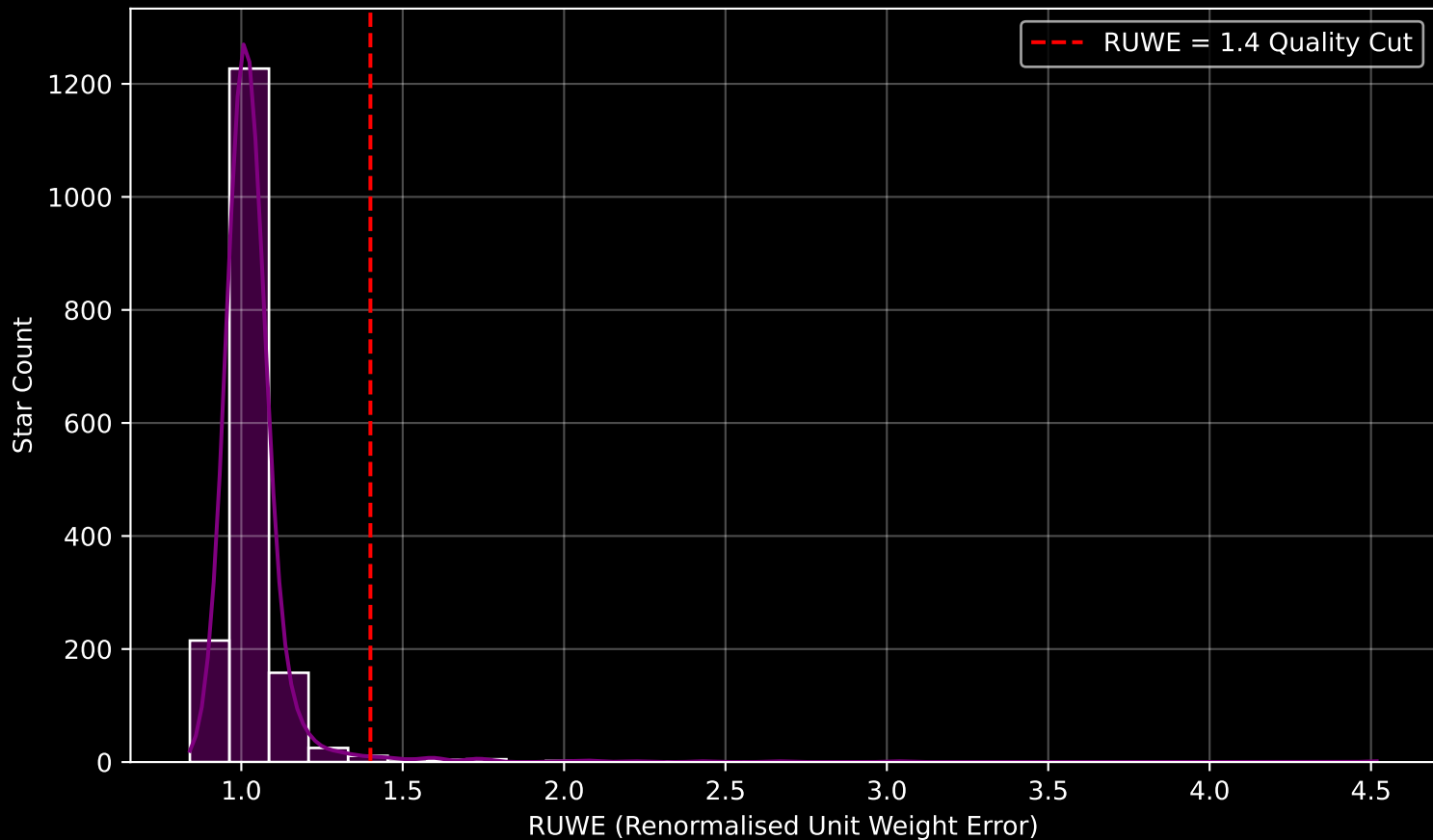
LAMOST DR11 Field: [GAC_080935N290534_M1] Parallax Quality (n= 1659)



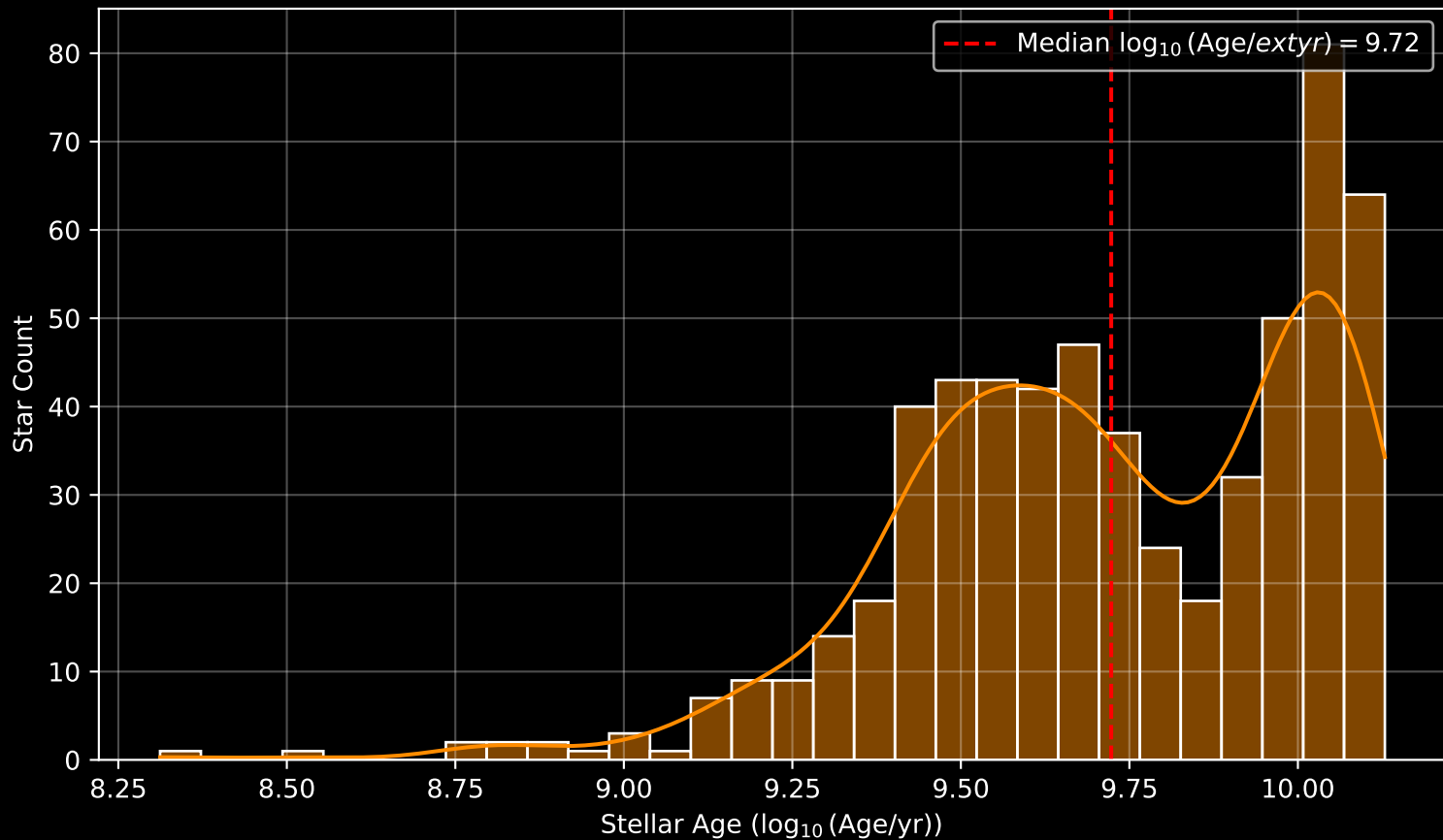
LAMOST DR11 Field: [GAC_080935N290534_M1] Stellar Population Parameters (n= 1659)



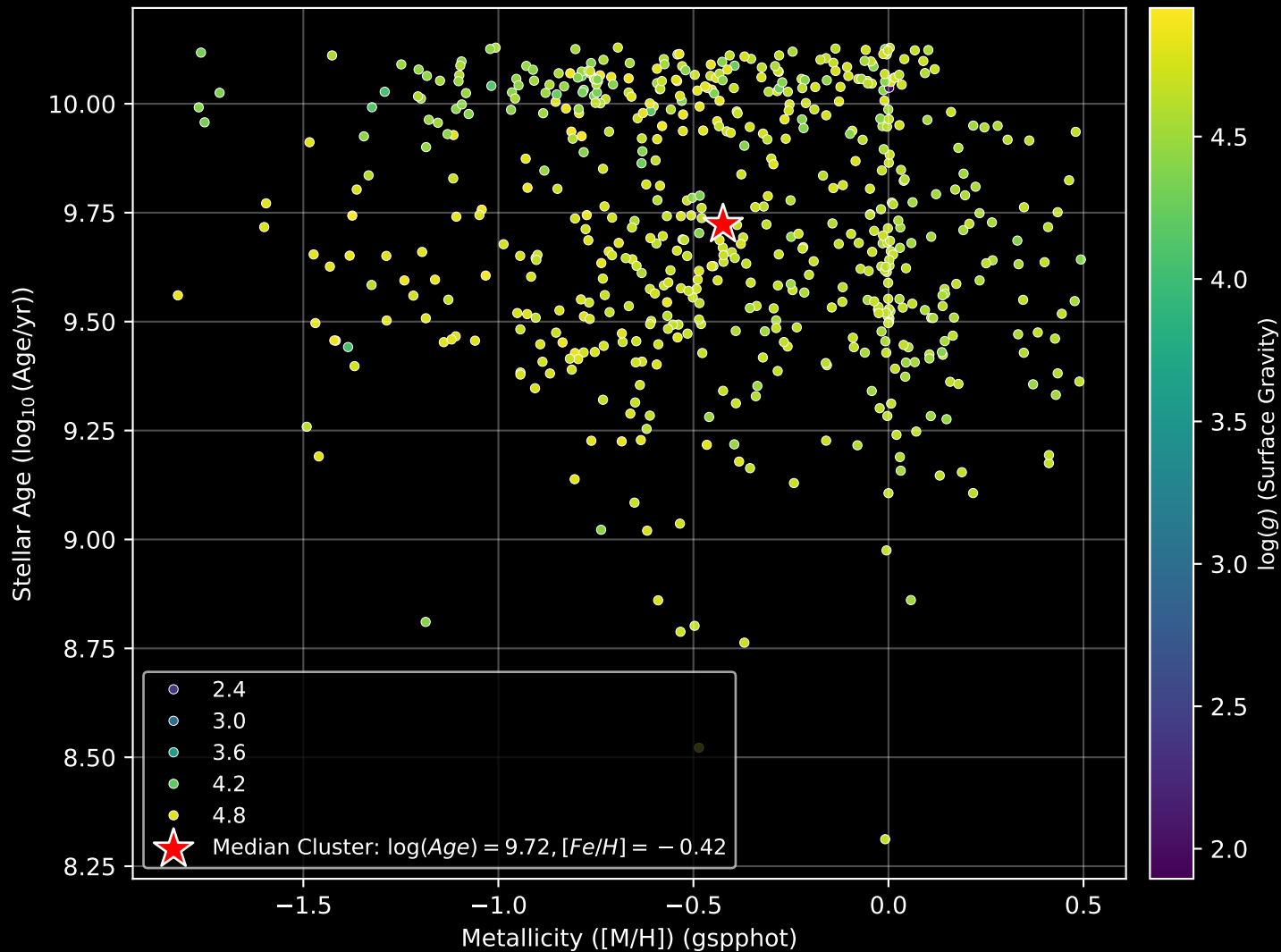
LAMOST DR11 Field [GAC_080935N290534_M1] RUWE Distribution (n= 1658)



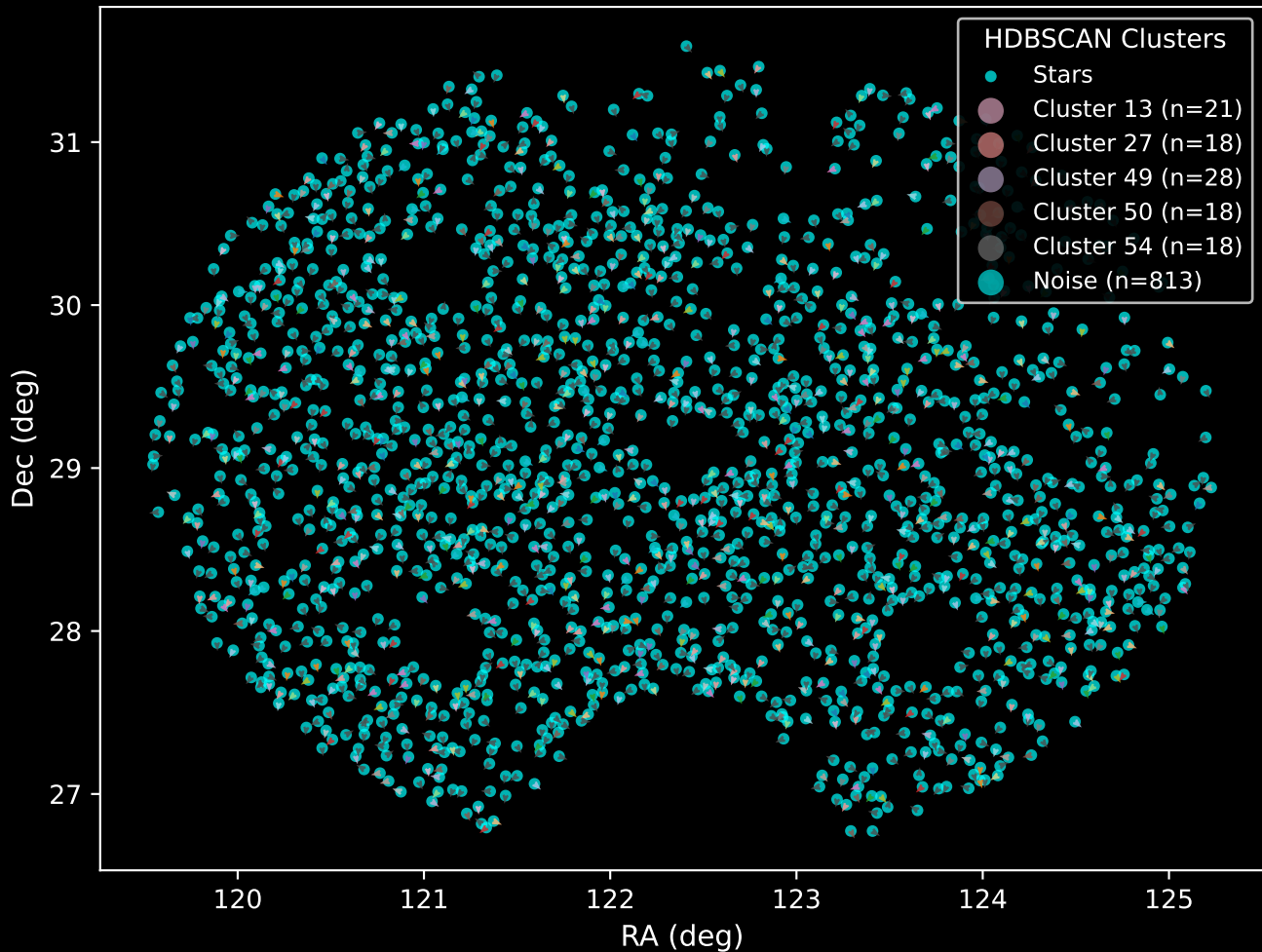
LAMOST DR11 Field: [GAC_080935N290534_M1] Age Distribution (n= 591)



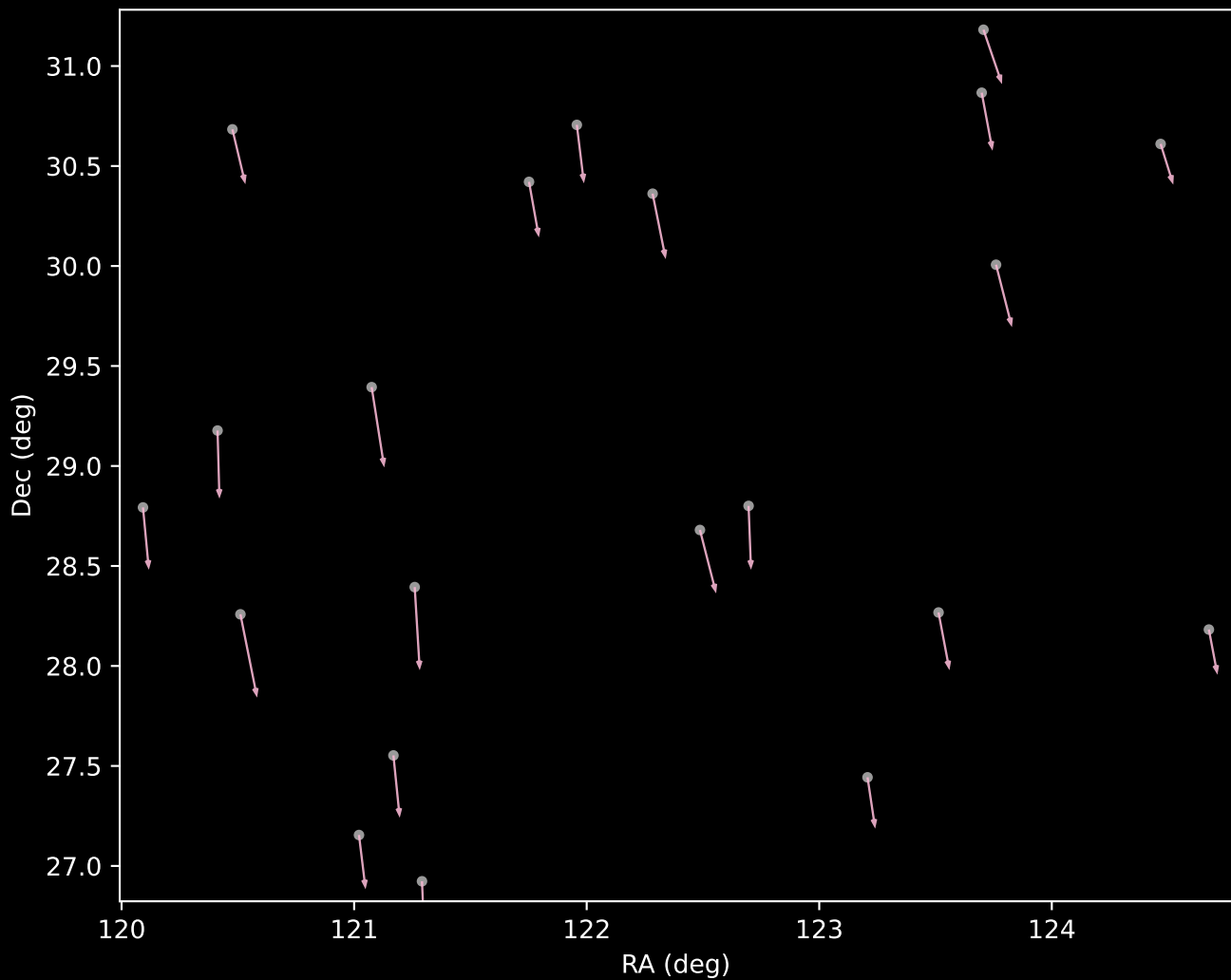
LAMOST DR11 Field: [GAC_080935N290534_M1] Age-Metallicity Relation (n= 591)



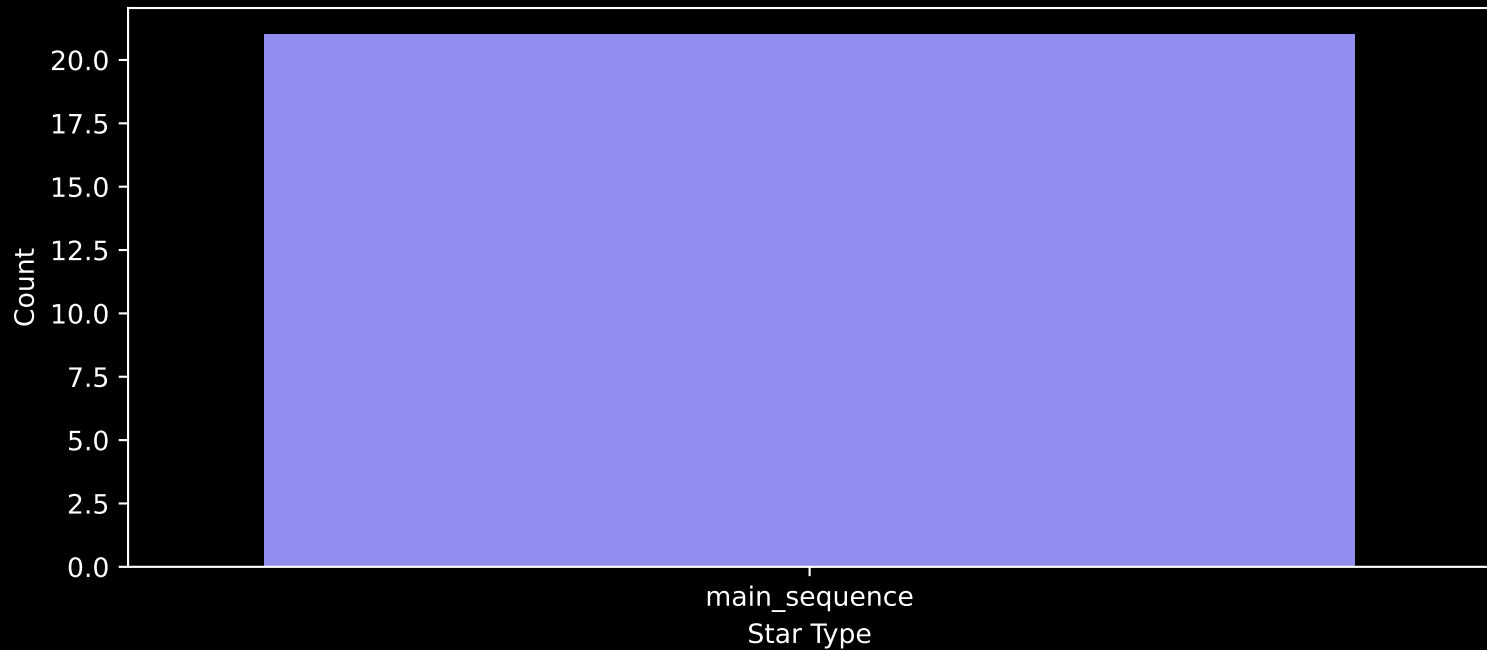
LAMOST DR11 Field [GAC_080935N290534_M1]
Proper Motion Vectors with HDBSCAN
(n=1659)



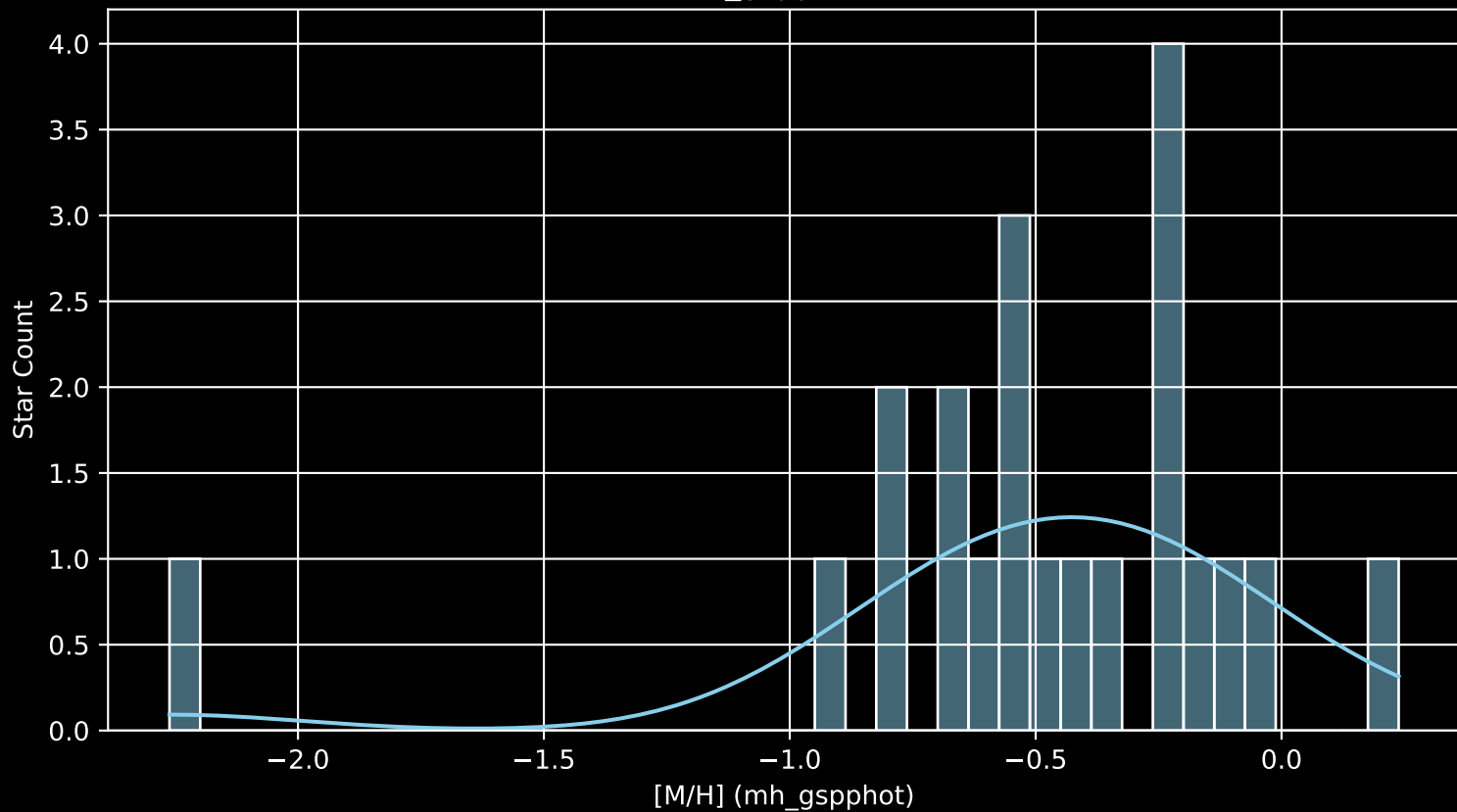
LAMOST DR11 Cluster 13 Proper Motion Vectors
(n=21)



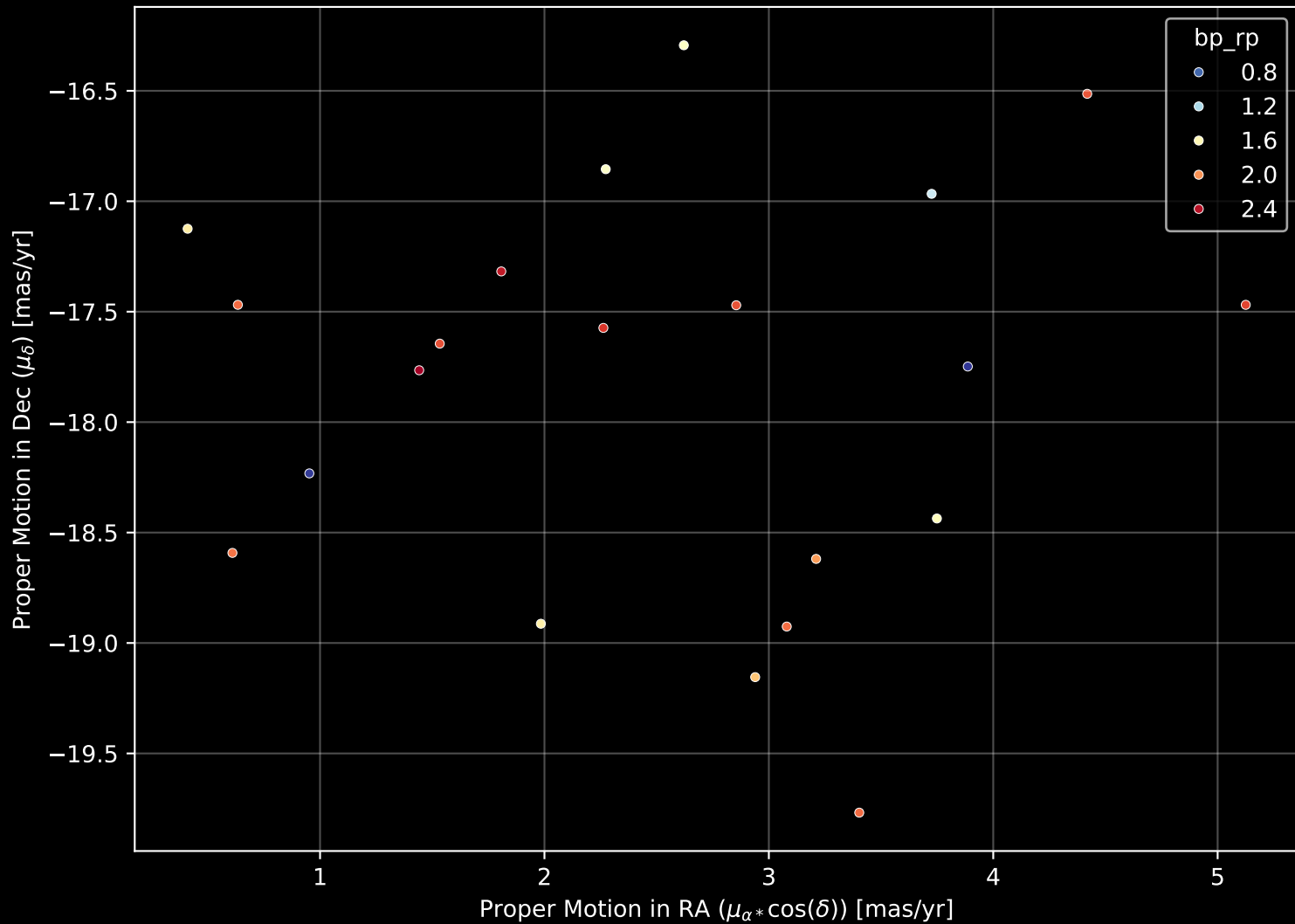
Cluster 13
Count plot of Star Type (n=21)



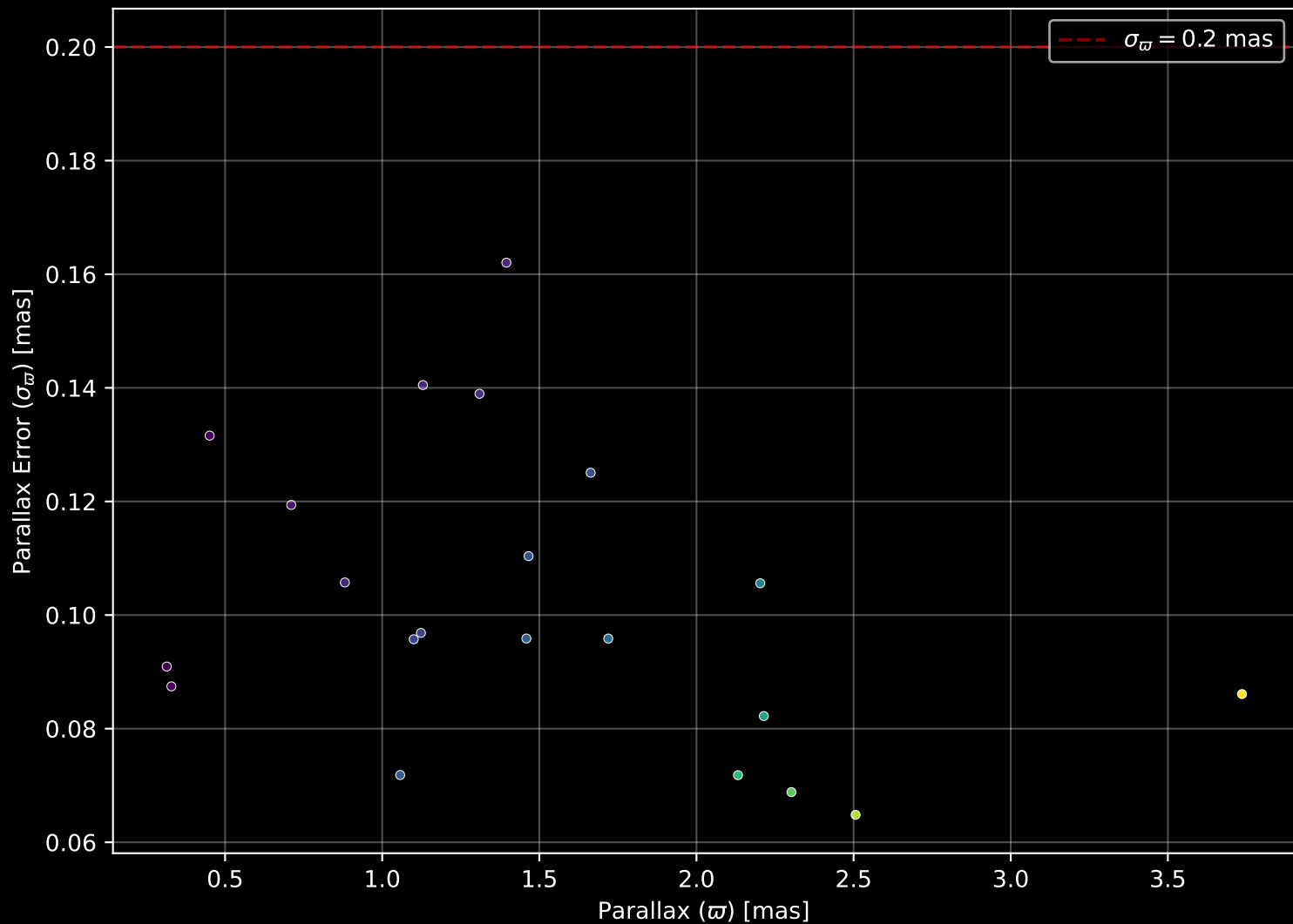
Cluster 13
[M/H] (mh_gspphot) (n=21)



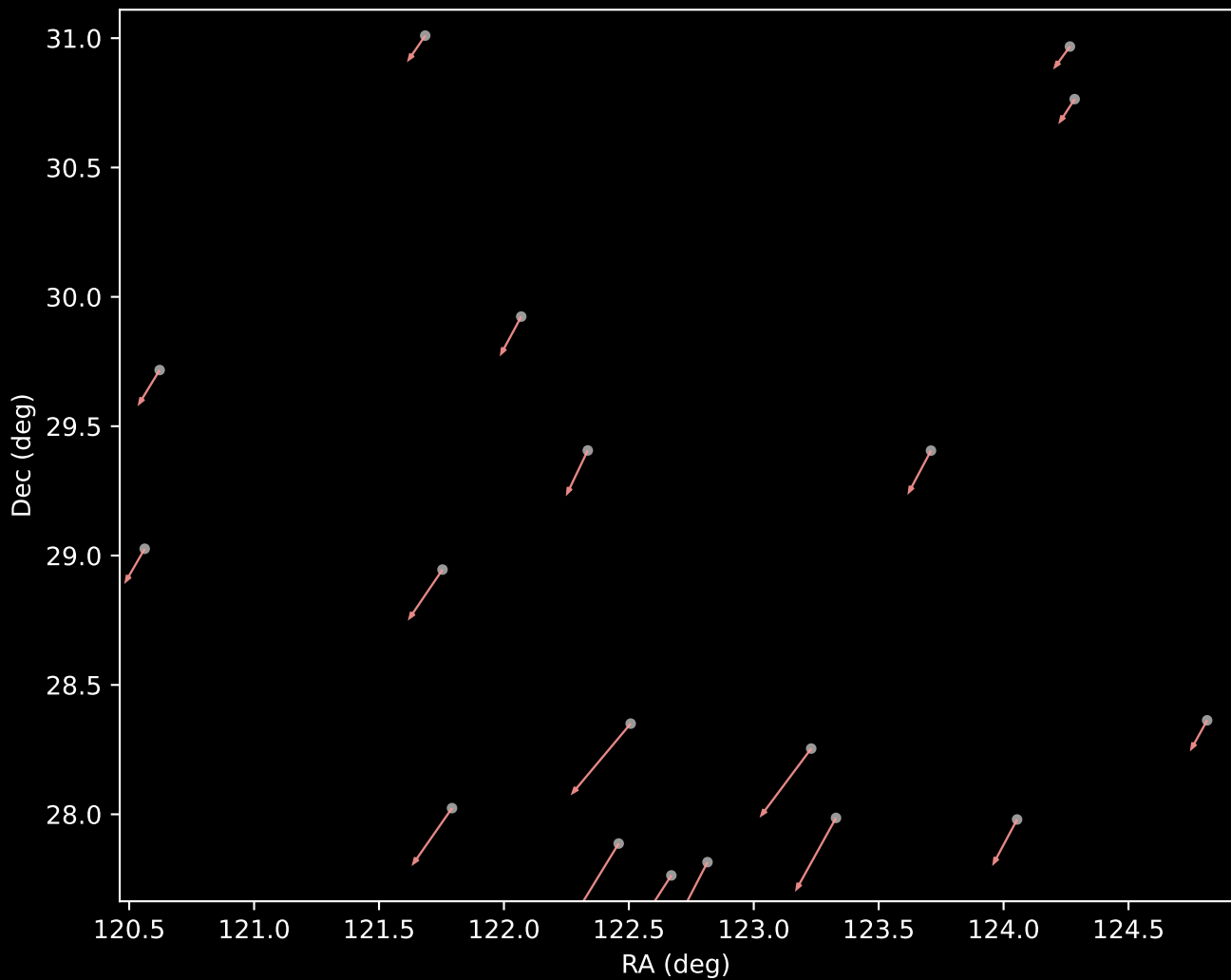
Cluster 13 Proper Motion Diagram (n=21)



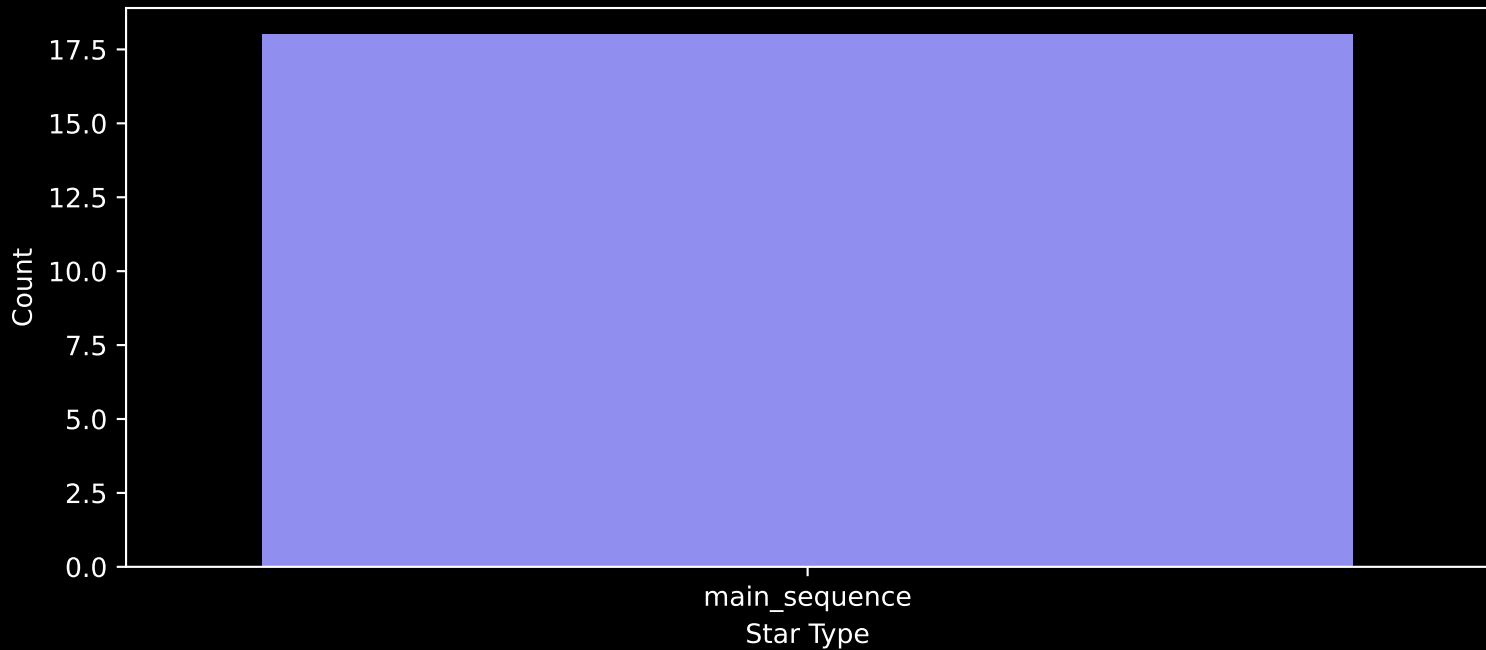
Cluster 13 Parallax Quality (n=21)



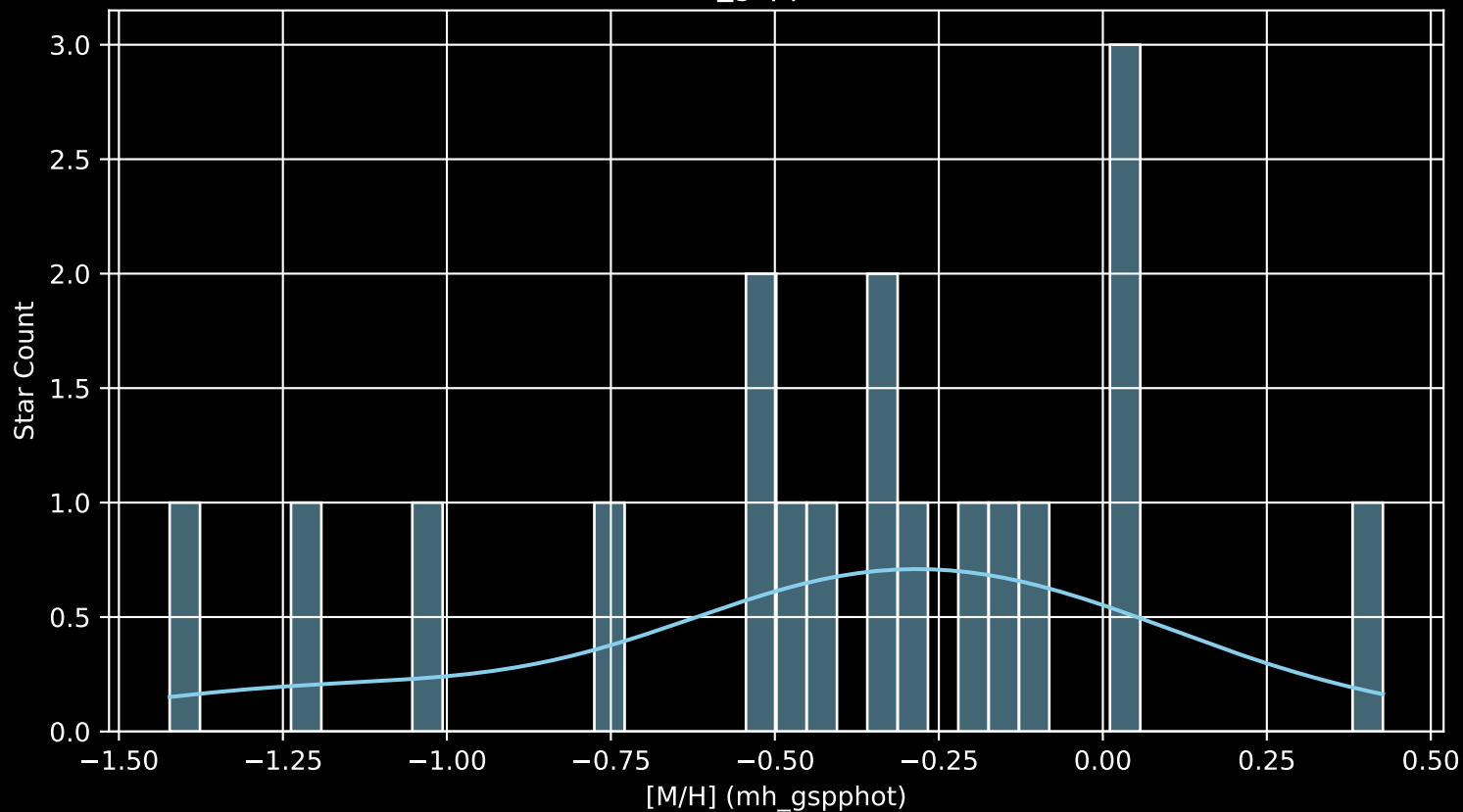
LAMOST DR11 Cluster 27 Proper Motion Vectors
(n=18)



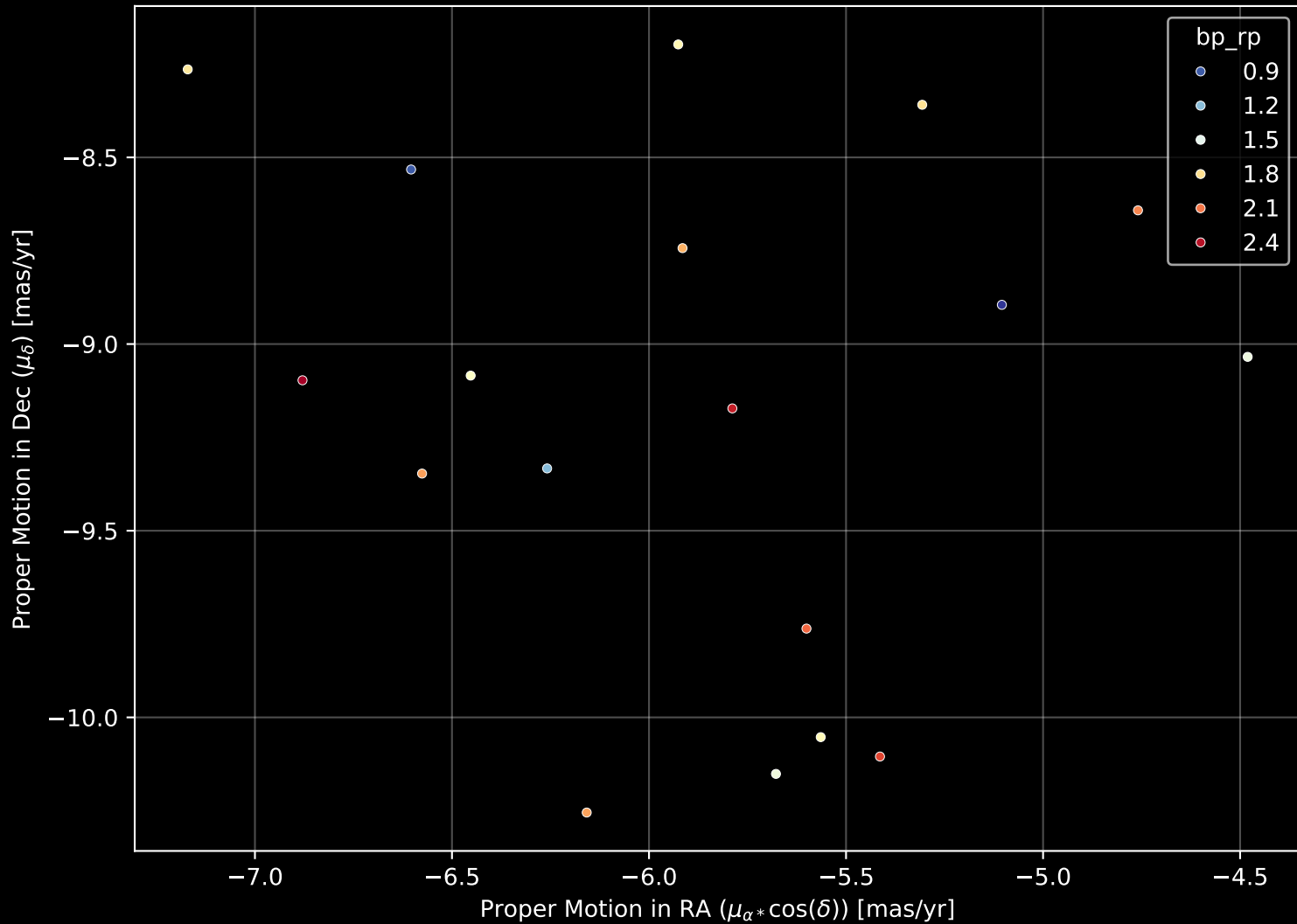
Cluster 27
Count plot of Star Type (n=18)



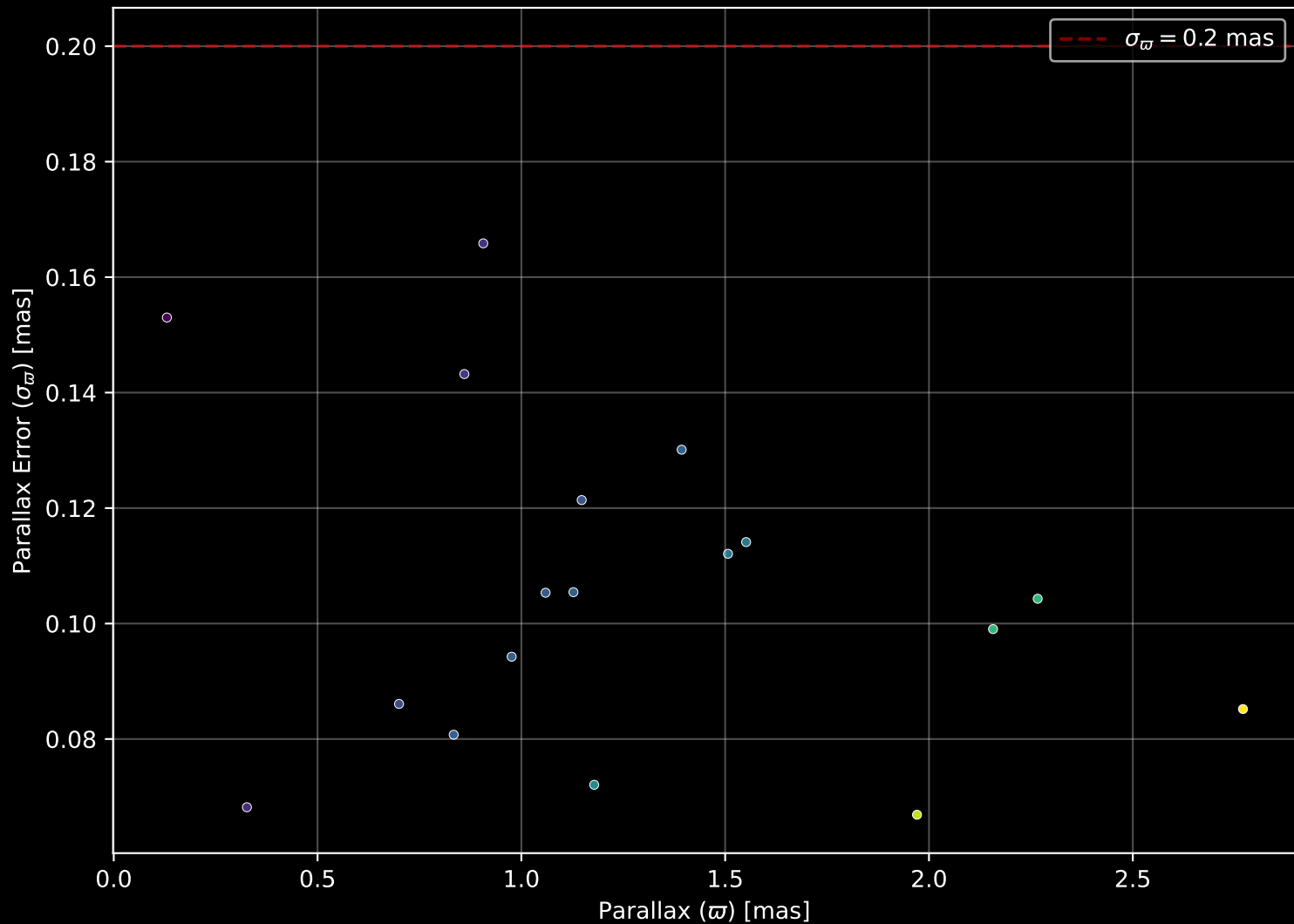
Cluster 27
[M/H] (mh_gspphot) (n=18)



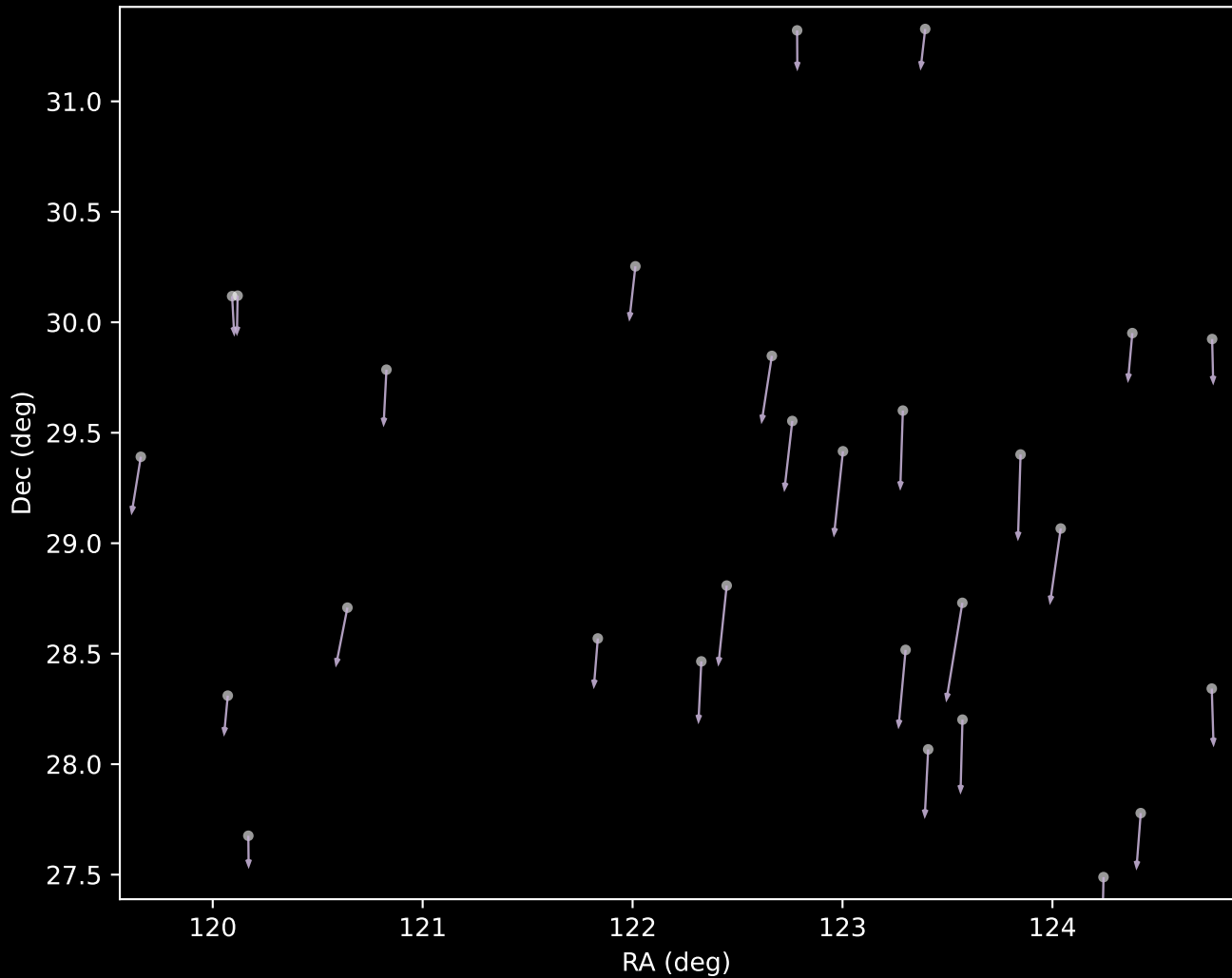
Cluster 27 Proper Motion Diagram (n=18)



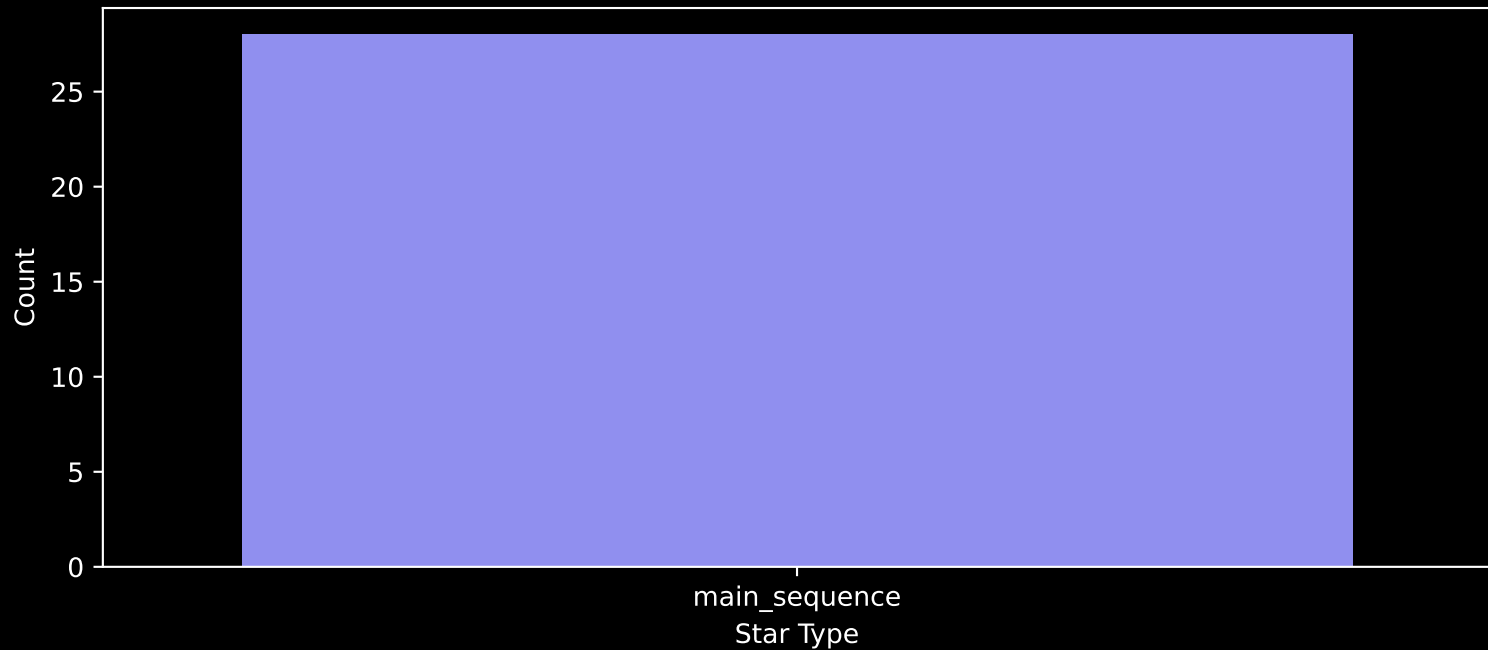
Cluster 27 Parallax Quality (n=18)



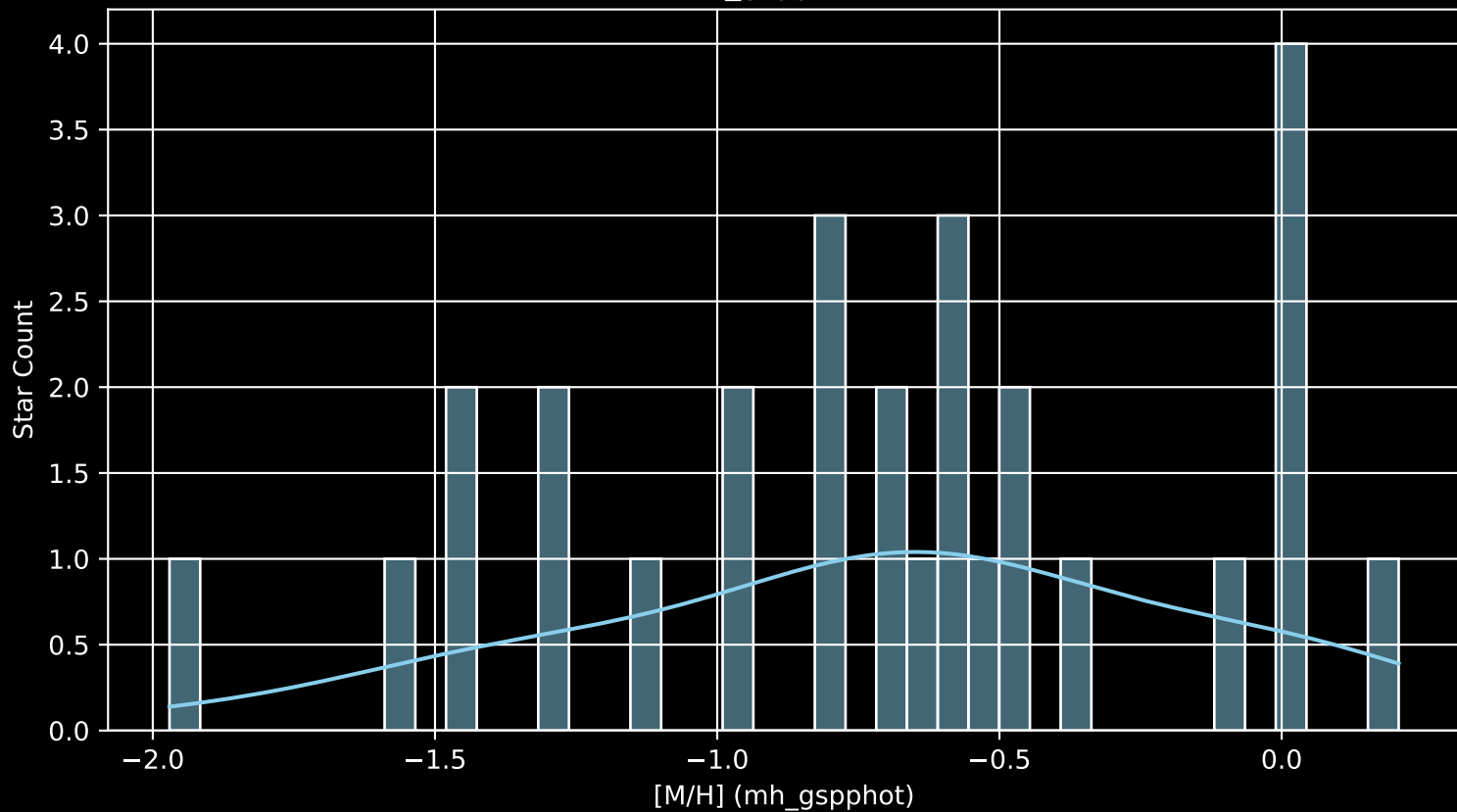
LAMOST DR11 Cluster 49 Proper Motion Vectors
(n=28)



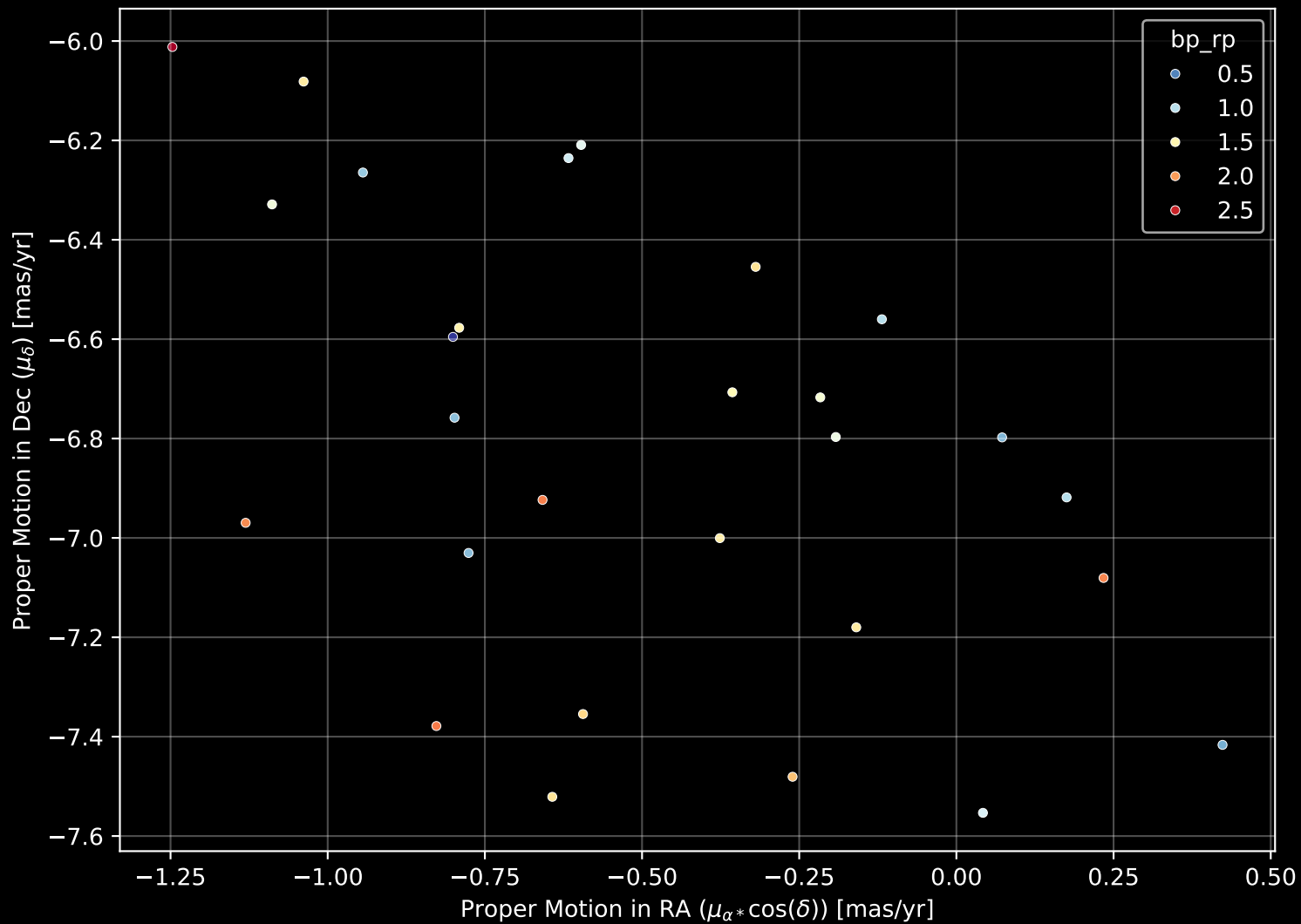
Cluster 49
Count plot of Star Type (n=28)



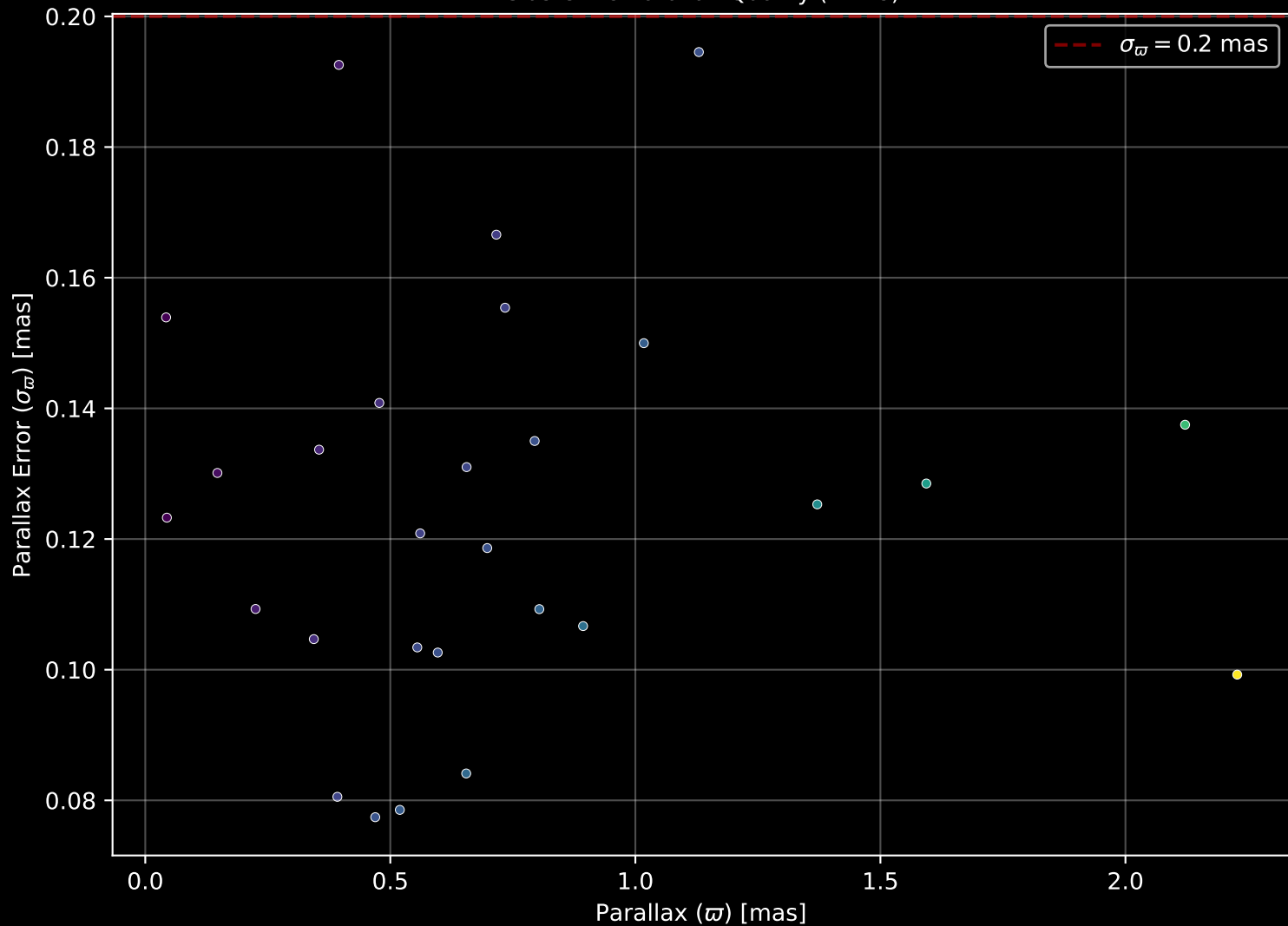
Cluster 49
[M/H] (mh_gspphot) (n=28)



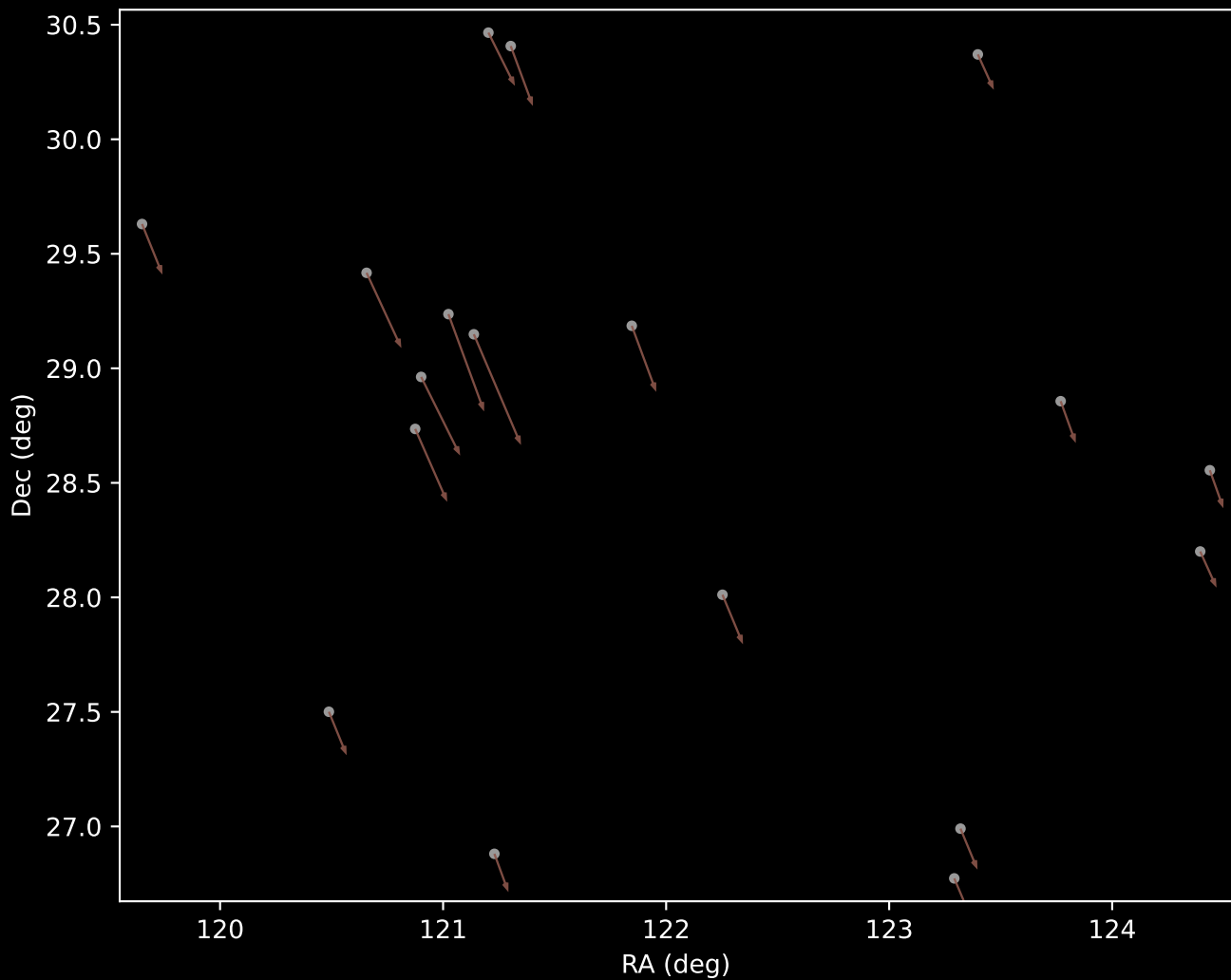
Cluster 49 Proper Motion Diagram (n=28)



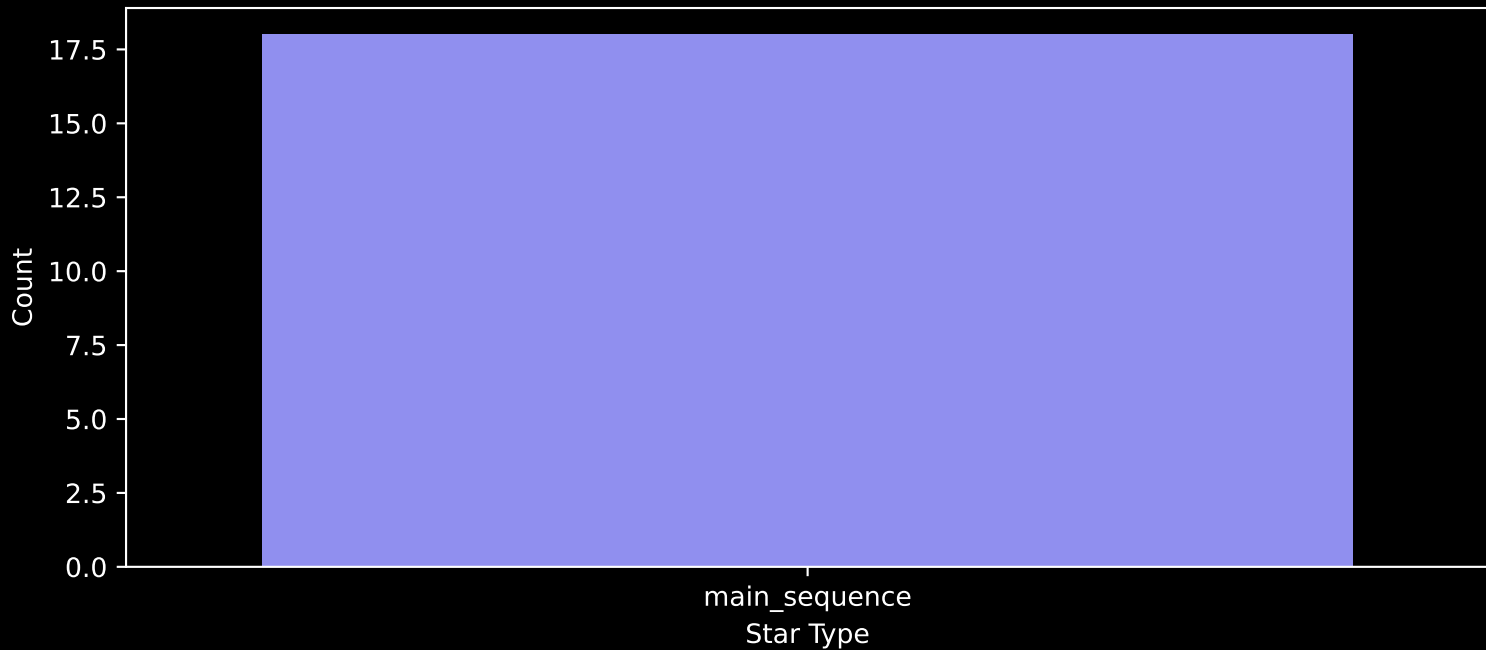
Cluster 49 Parallax Quality (n=28)



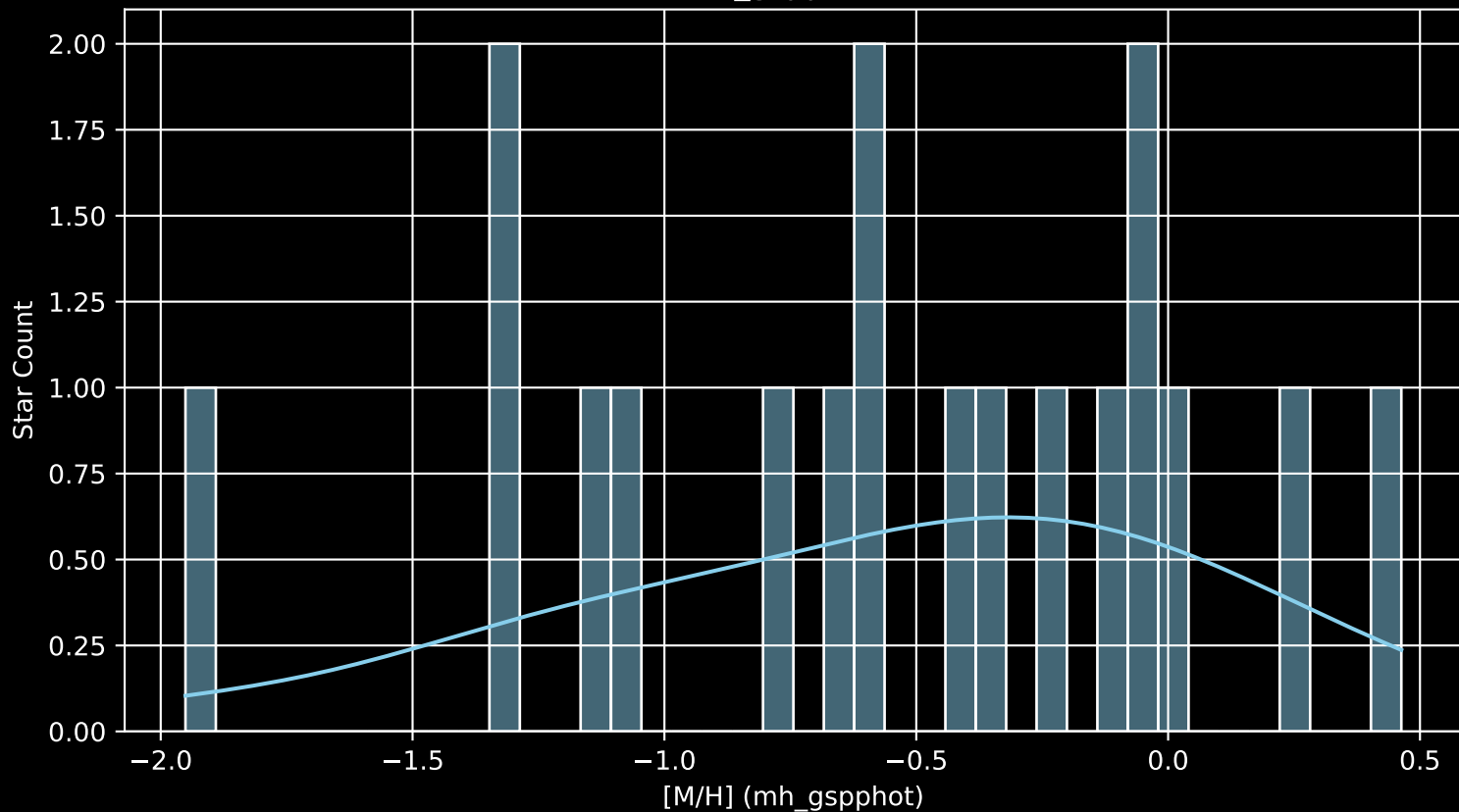
LAMOST DR11 Cluster 50 Proper Motion Vectors
(n=18)



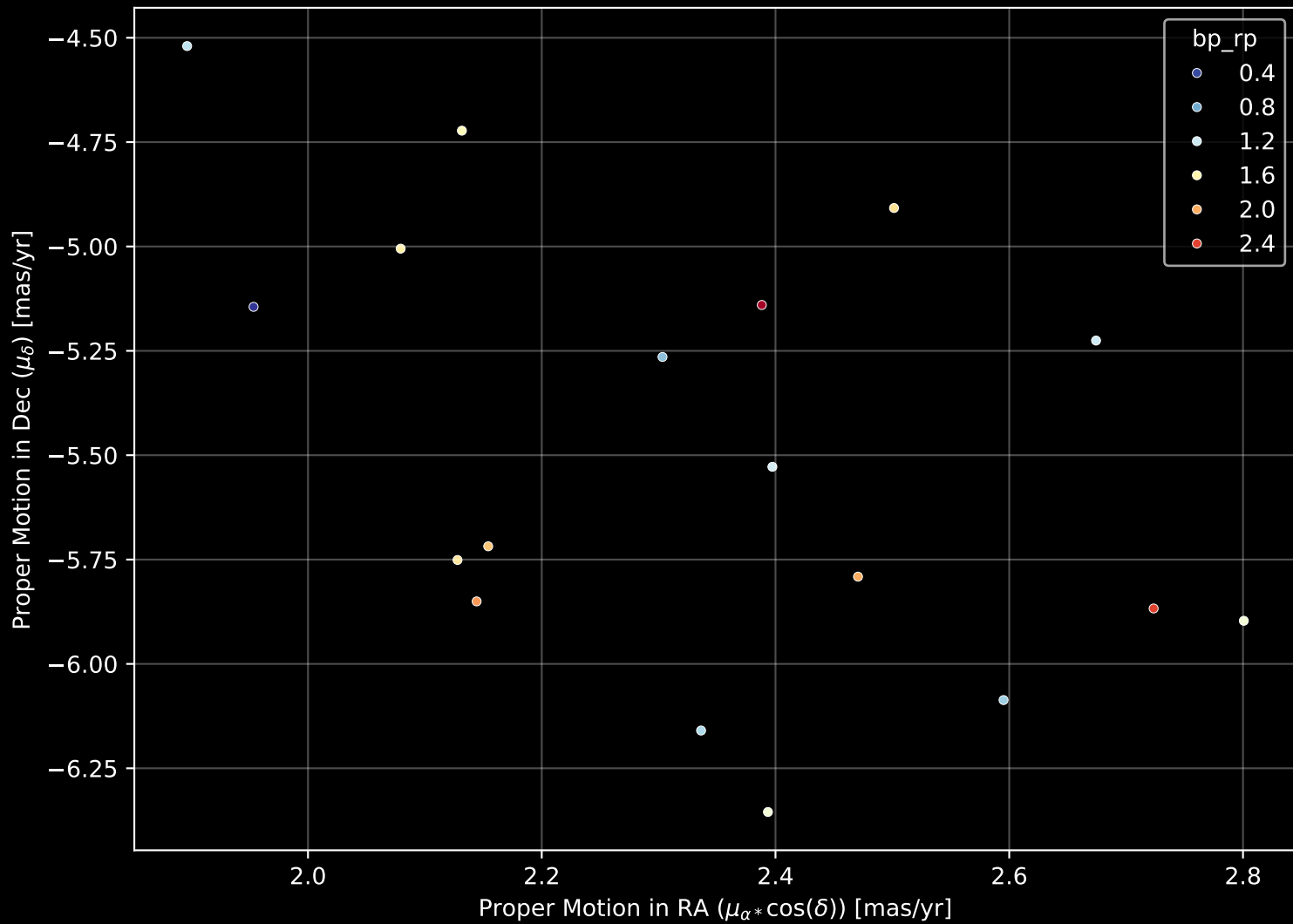
Cluster 50
Count plot of Star Type (n=18)



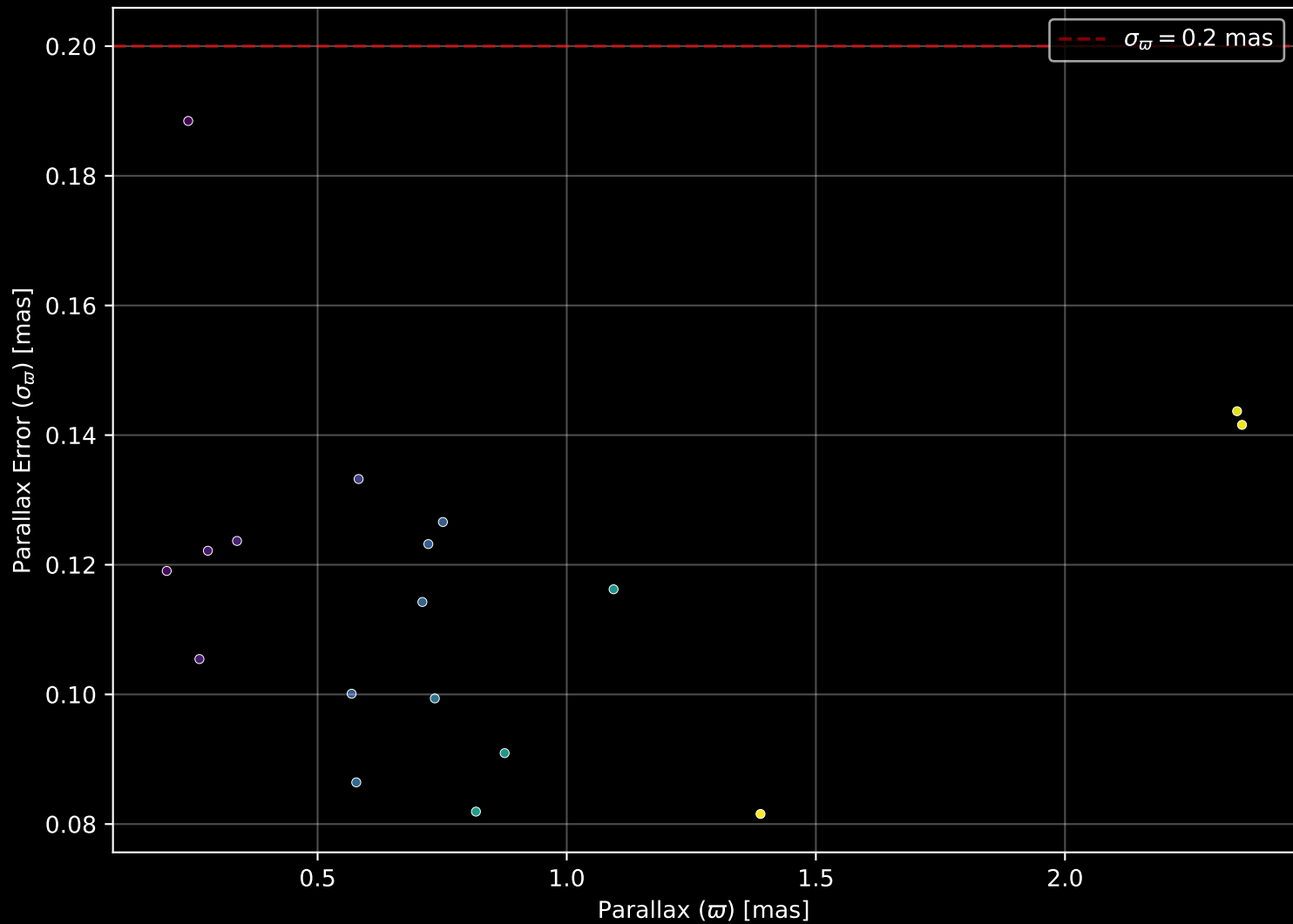
Cluster 50
[M/H] (mh_gspphot) (n=18)



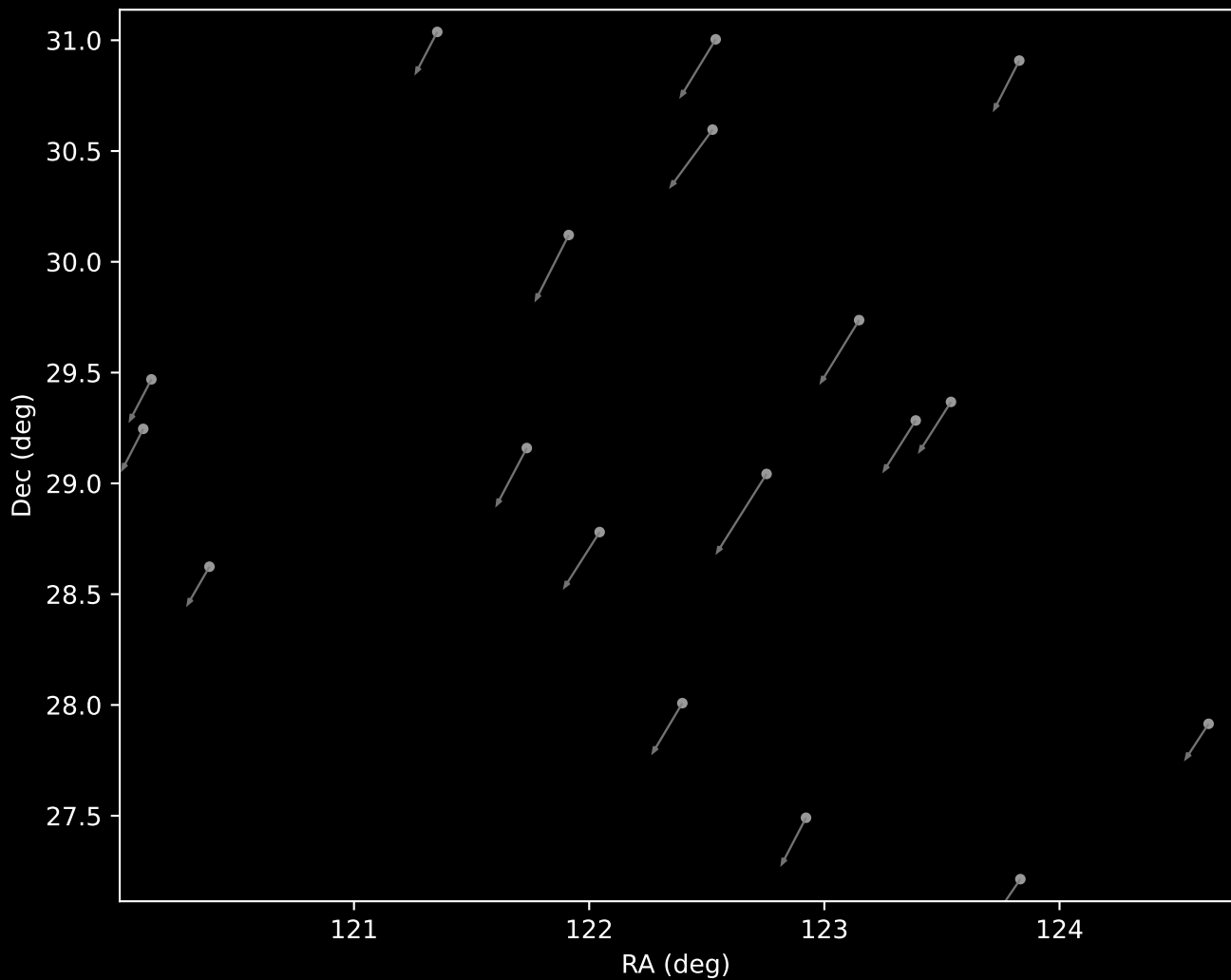
Cluster 50 Proper Motion Diagram (n=18)



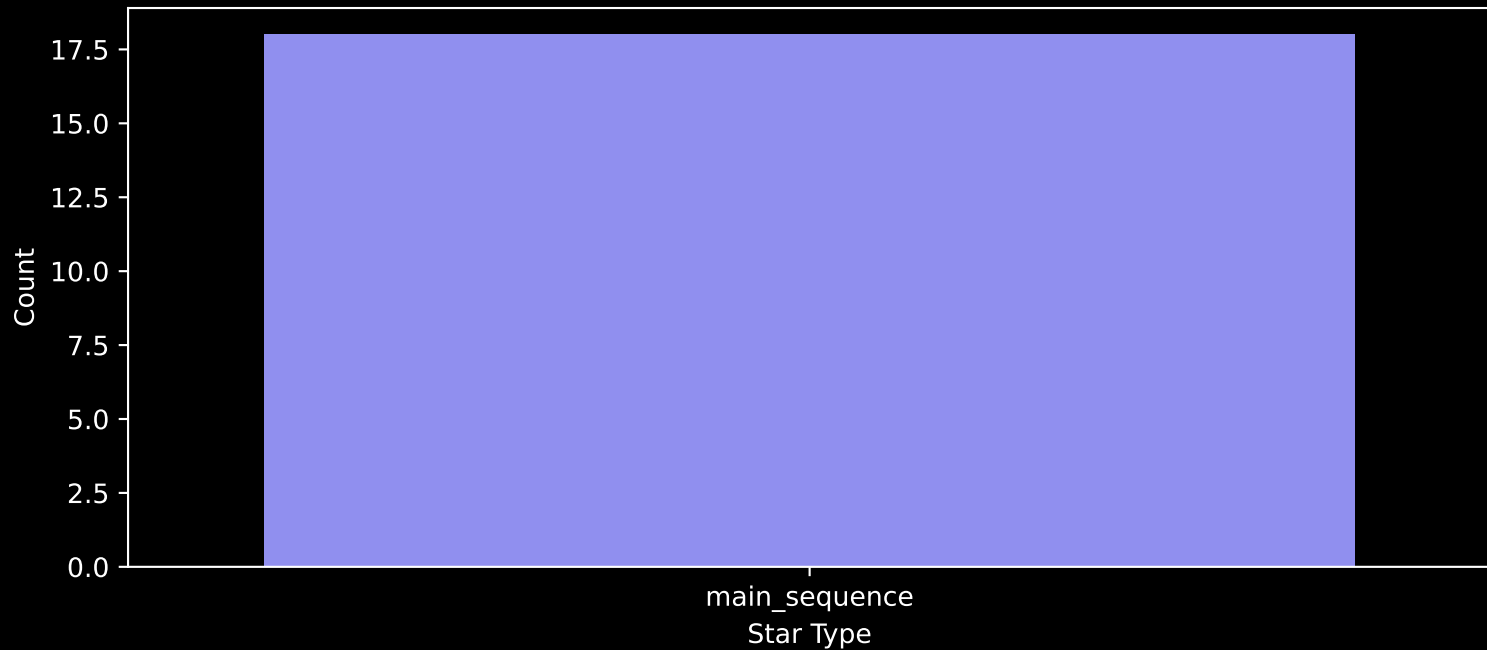
Cluster 50 Parallax Quality (n=18)



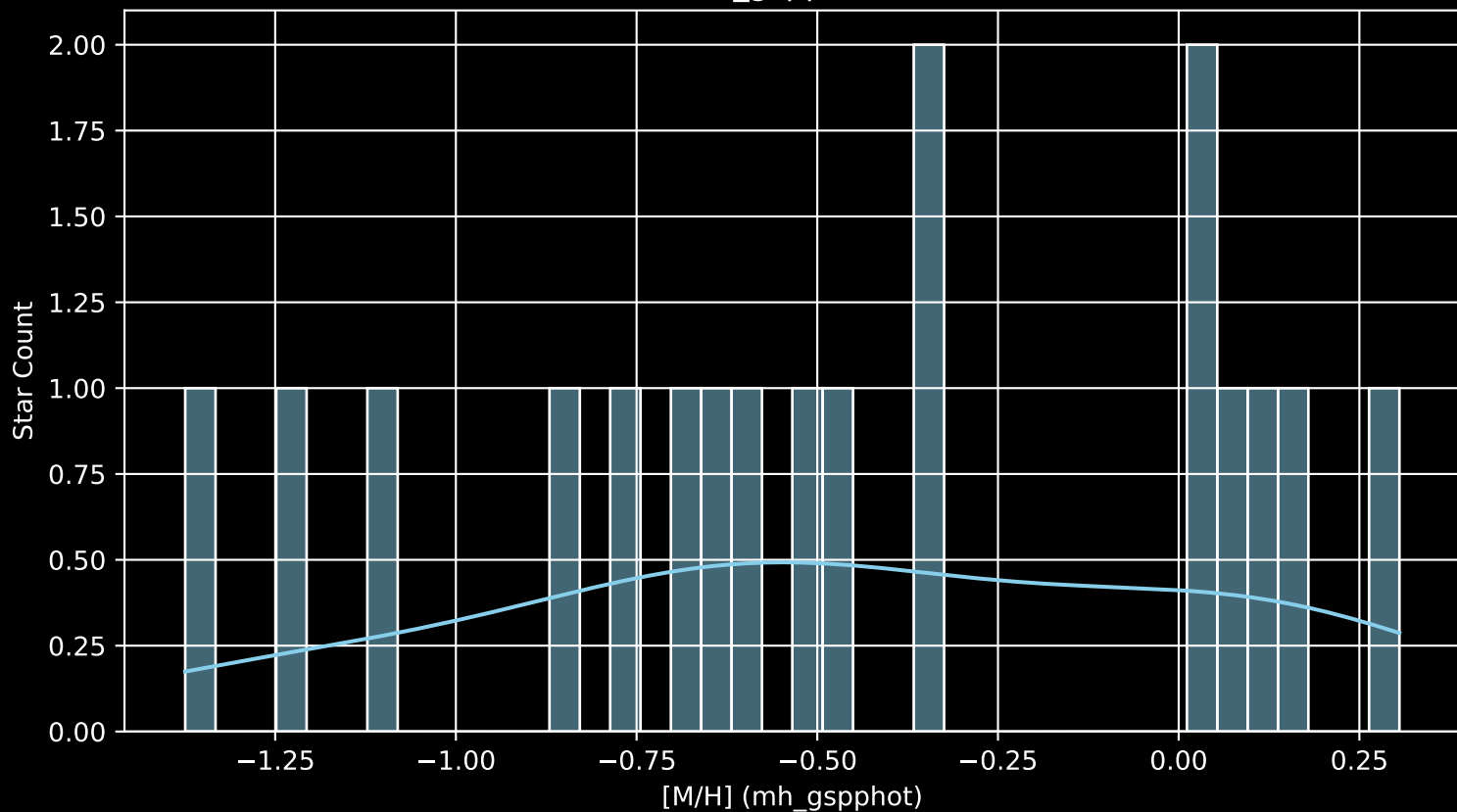
LAMOST DR11 Cluster 54 Proper Motion Vectors
(n=18)



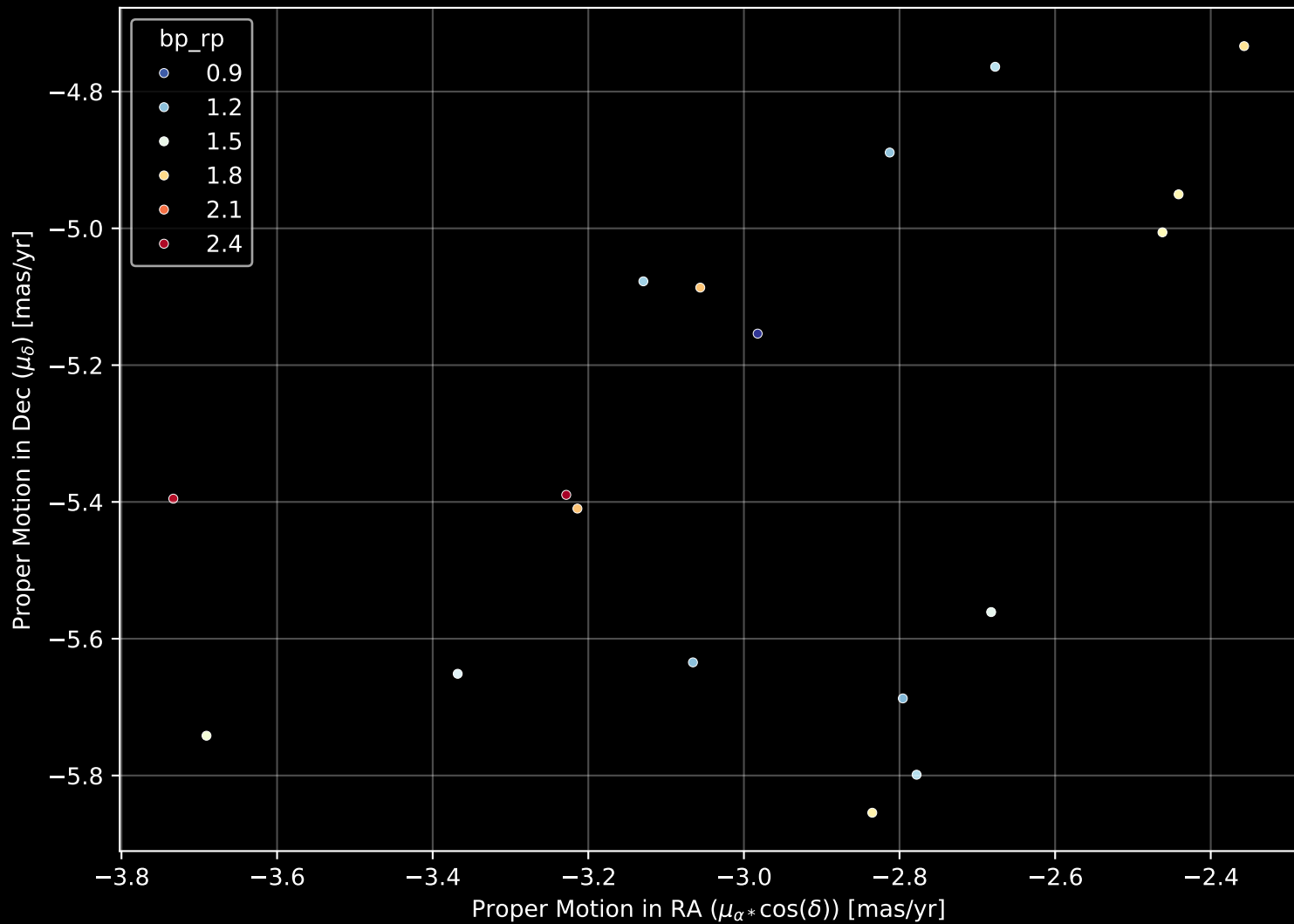
Cluster 54
Count plot of Star Type (n=18)



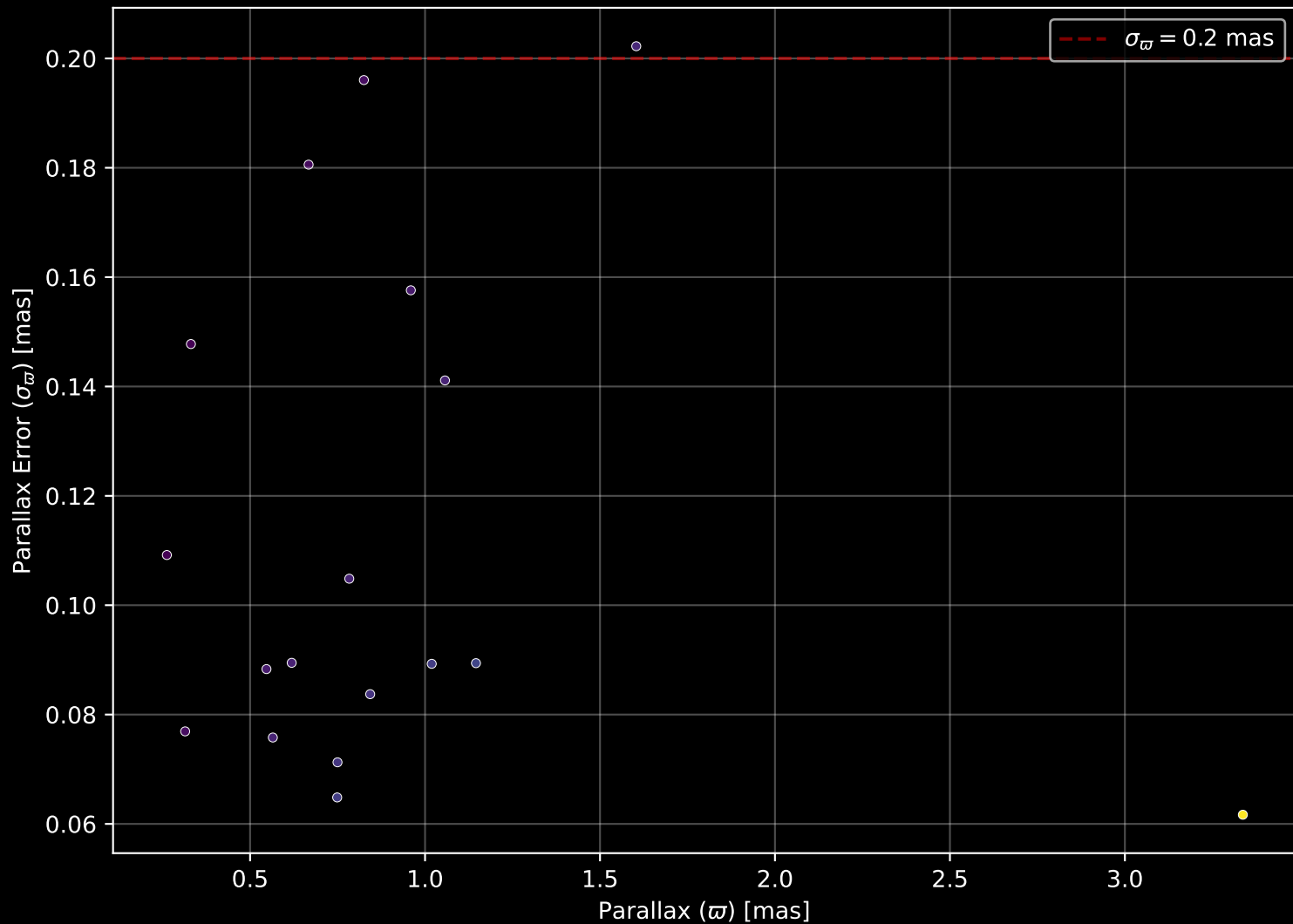
Cluster 54
[M/H] (mh_gspphot) (n=18)



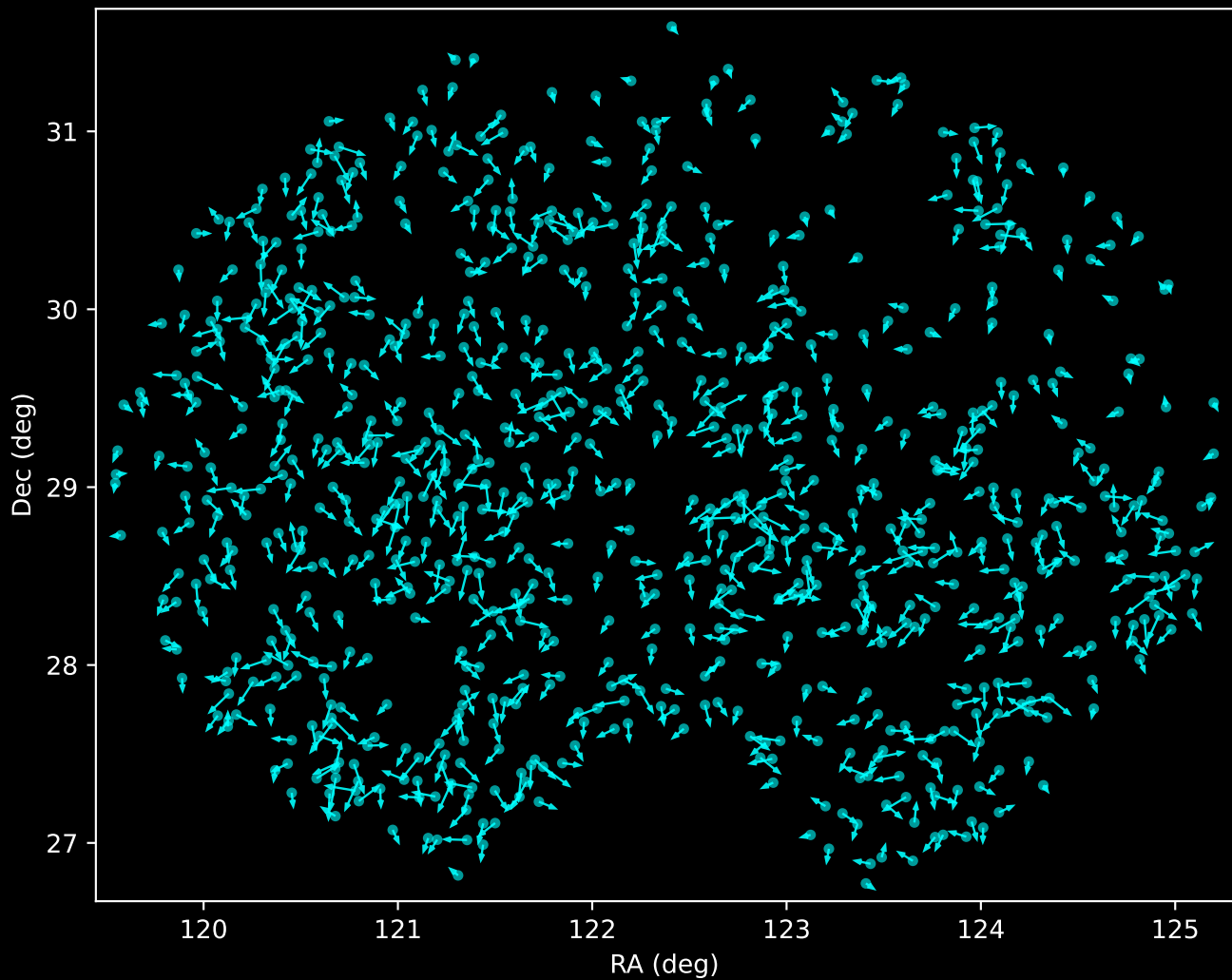
Cluster 54 Proper Motion Diagram (n=18)



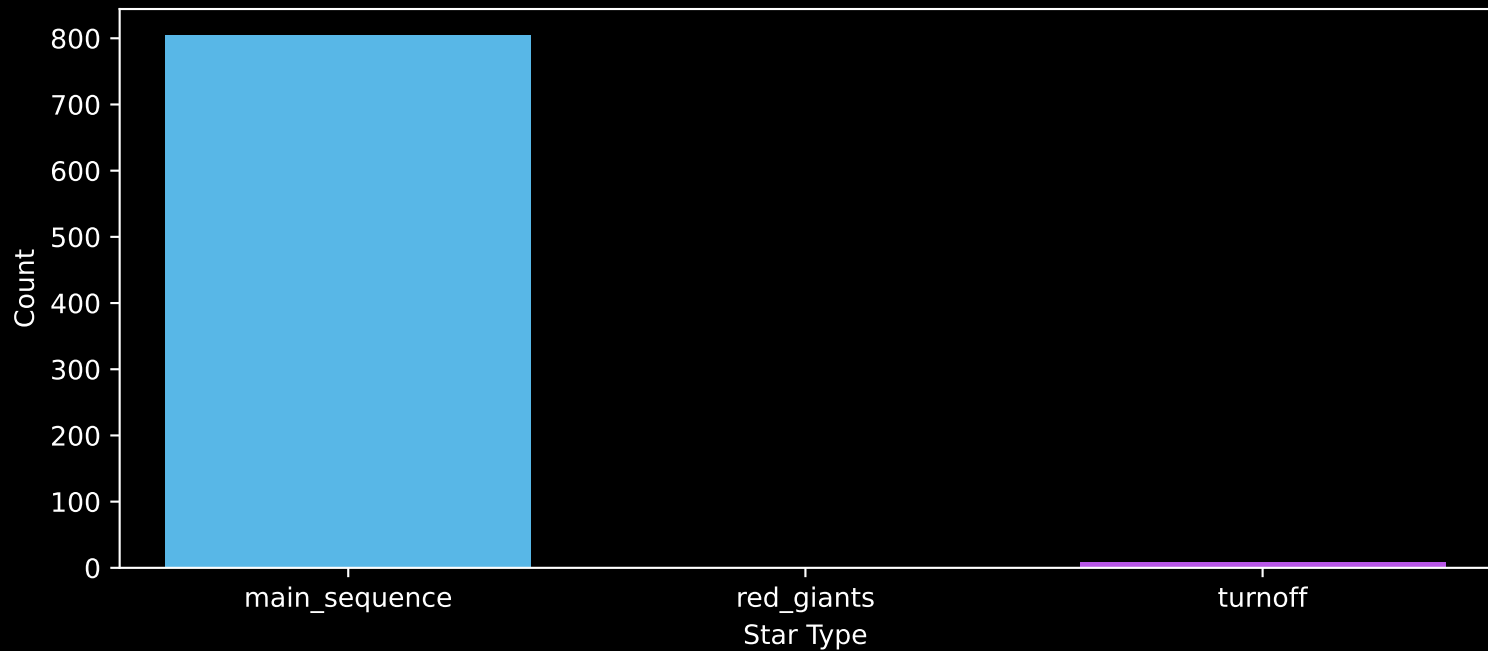
Cluster 54 Parallax Quality (n=18)



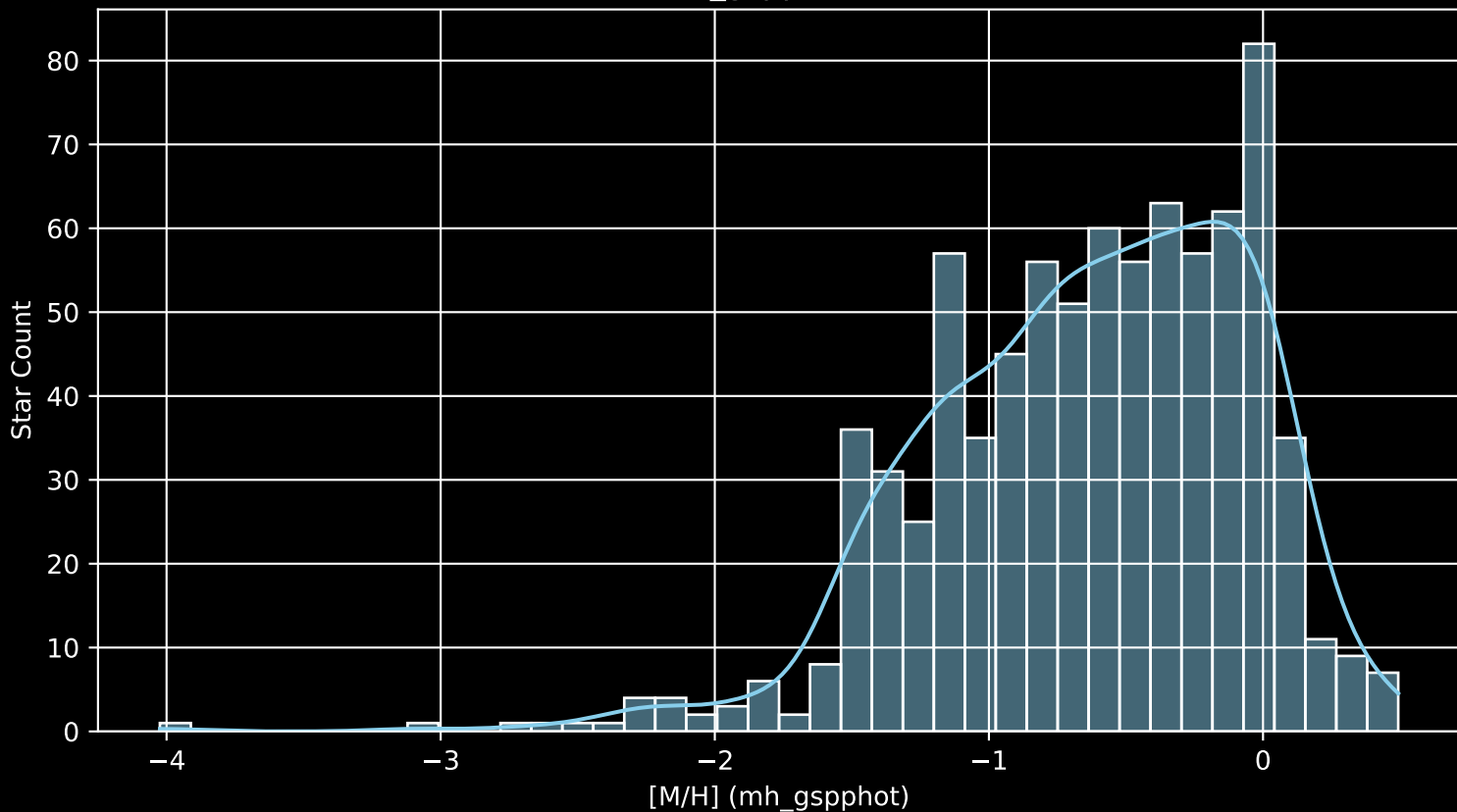
LAMOST DR11 Noise Stream Proper Motion Vectors
(n=813)



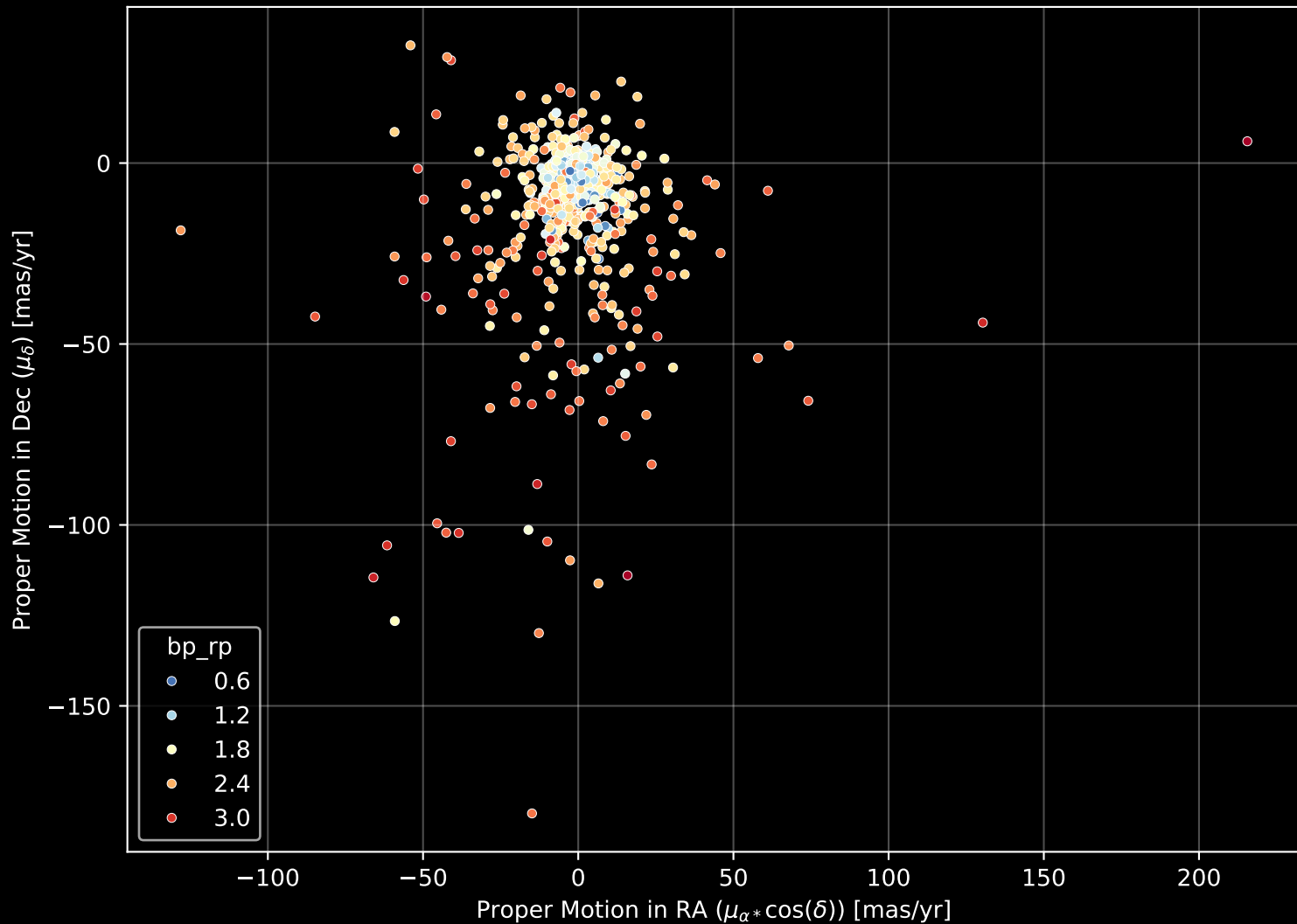
Cluster -1
Count plot of Star Type (n=813)



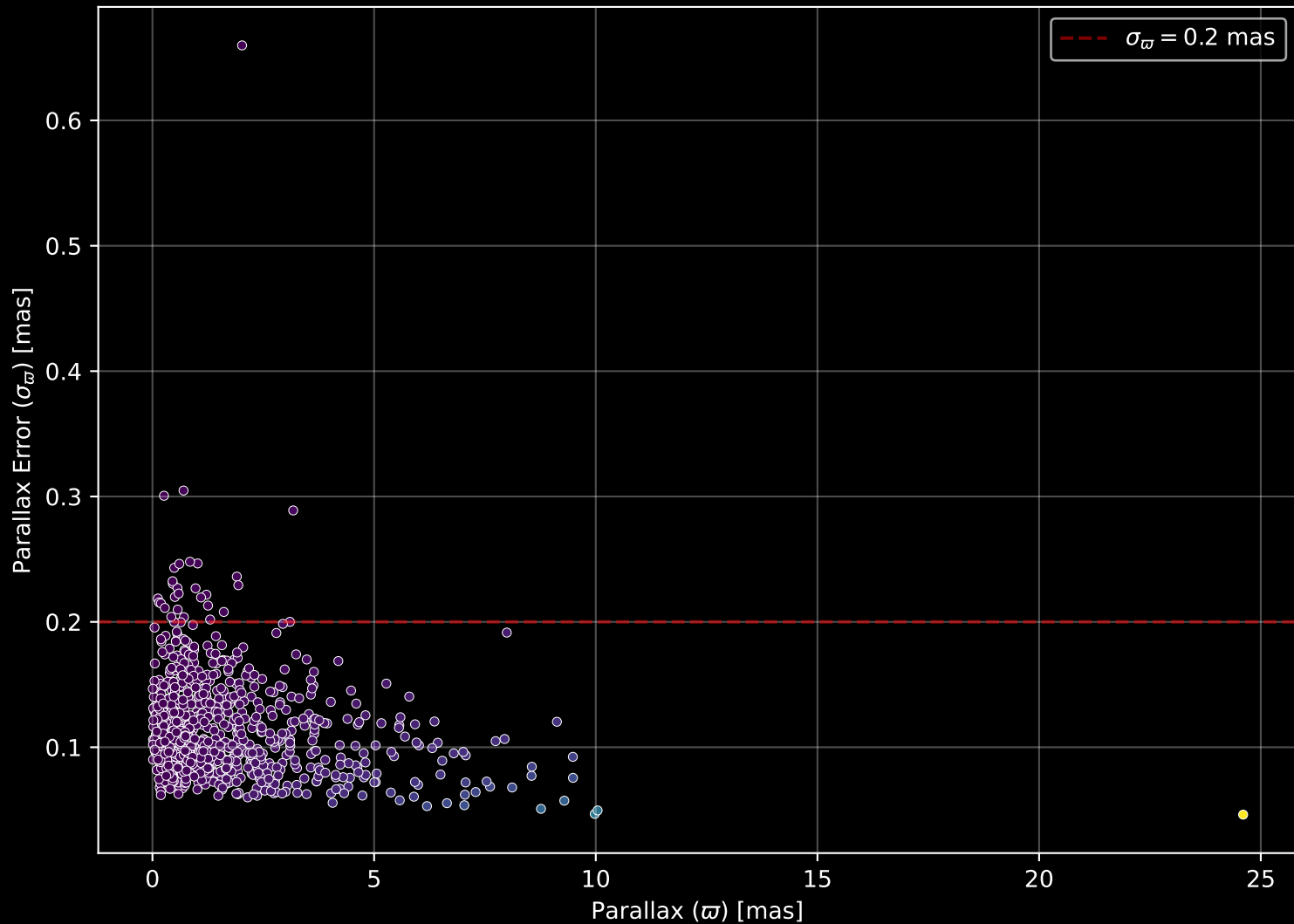
Cluster -1
[M/H] (mh_gspphot) (n=813)



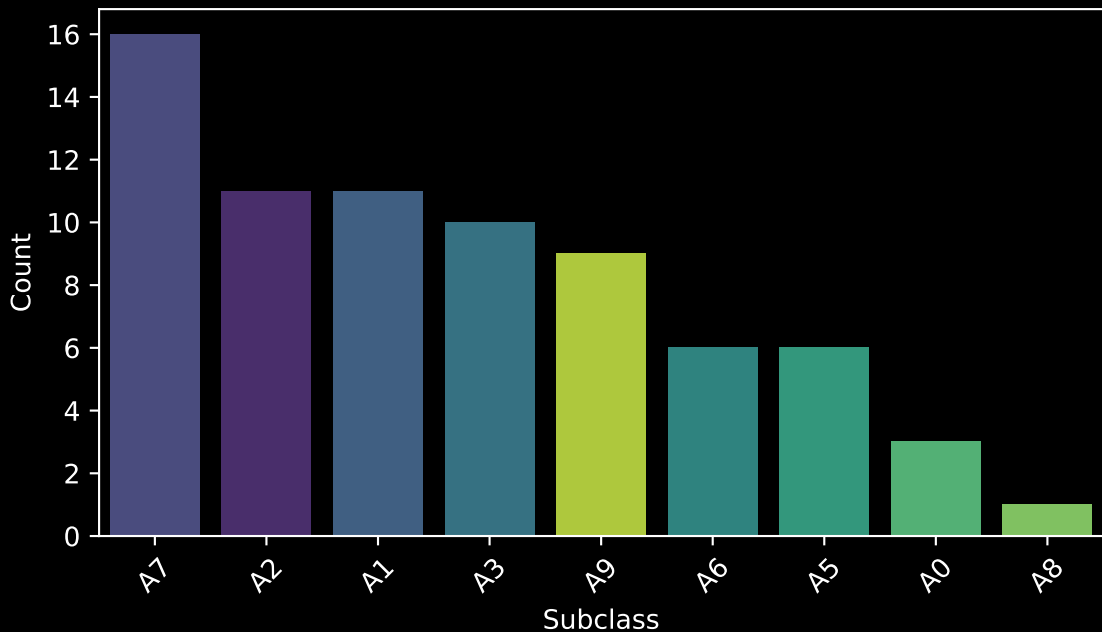
Cluster -1 Proper Motion Diagram (n=813)



Cluster -1 Parallax Quality (n=813)

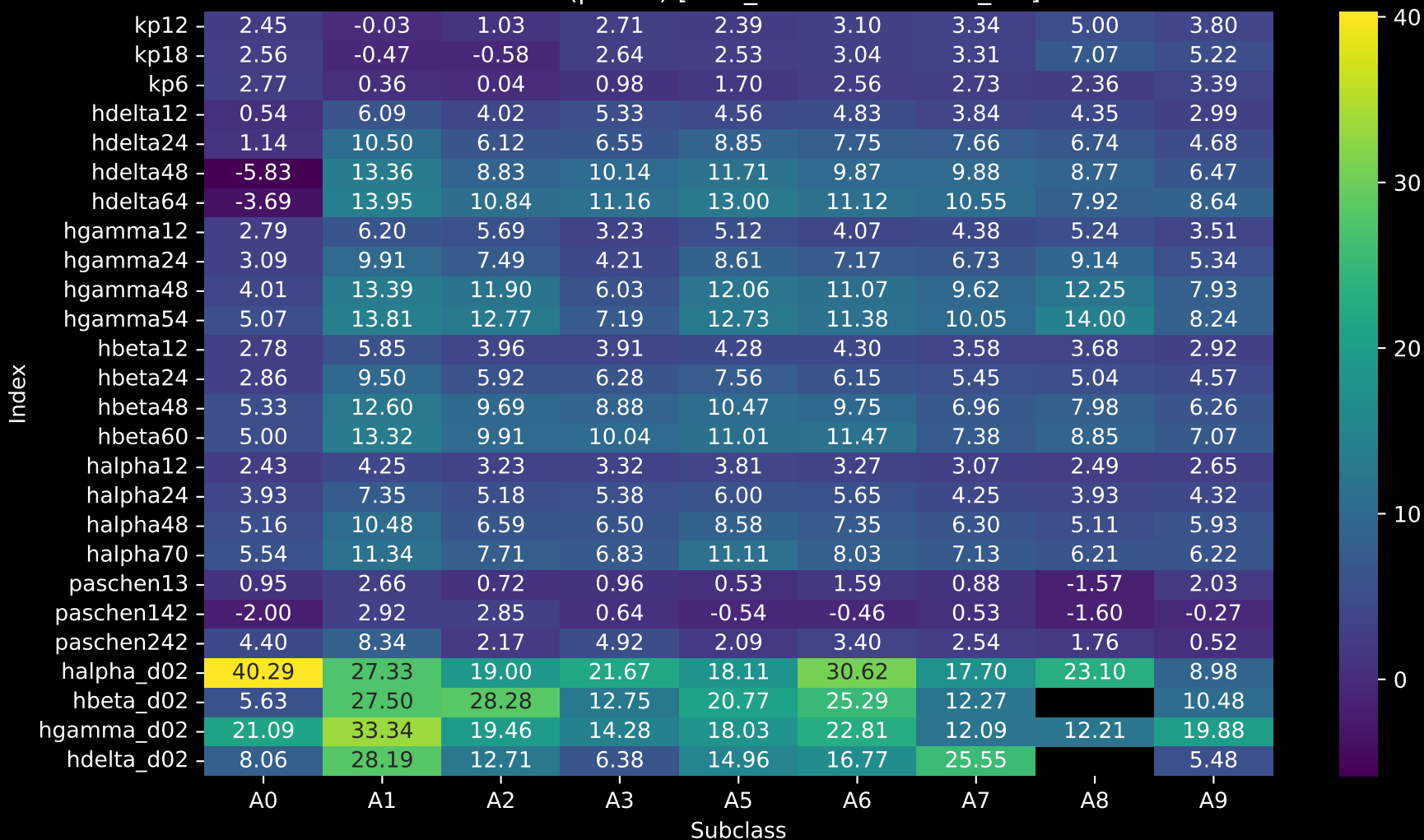


Subclass distribution for planid GAC_080935N290534_M1 (n= 73)

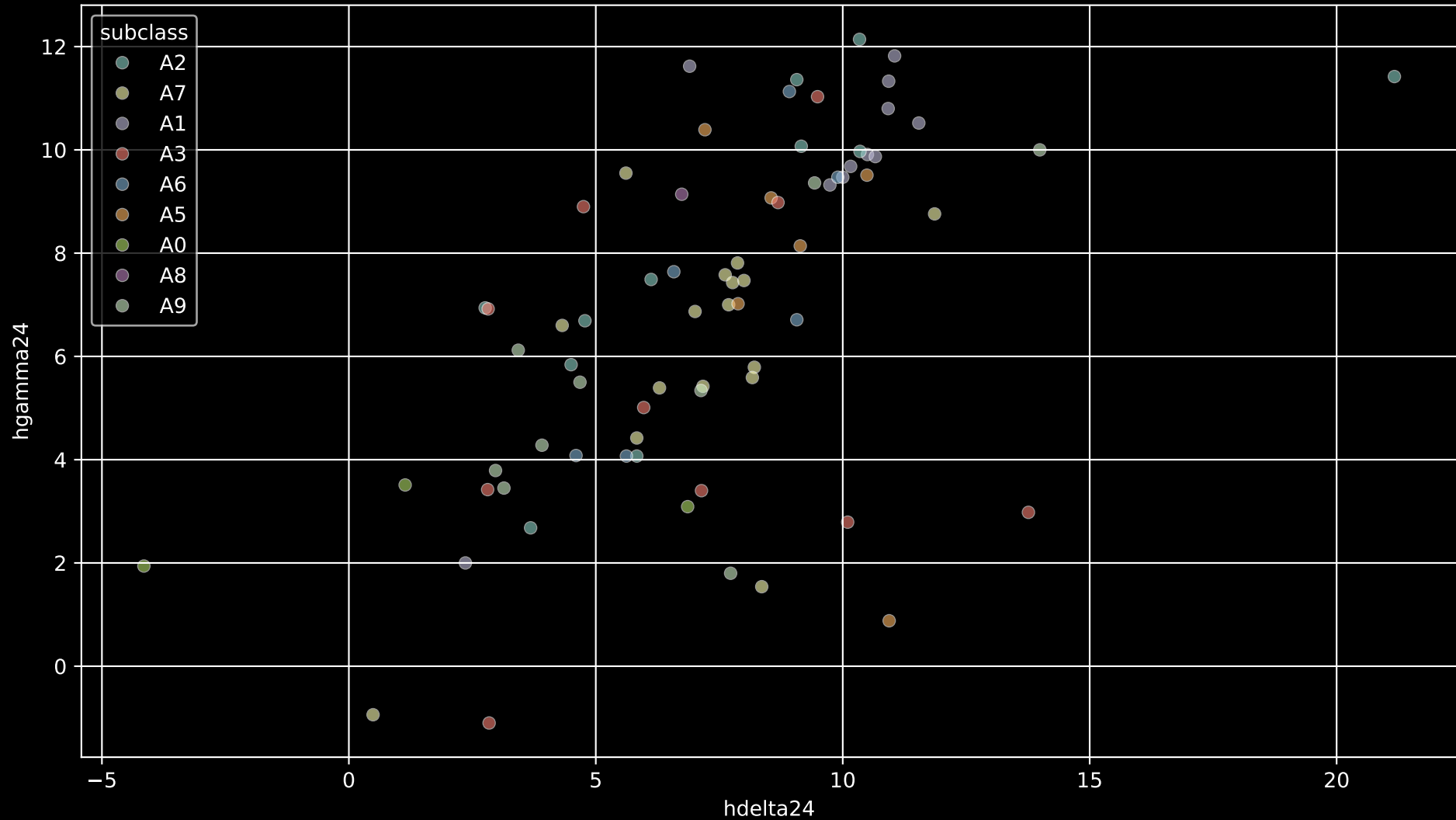


LAMOST DR11 Median Spectra Indices By Subclass

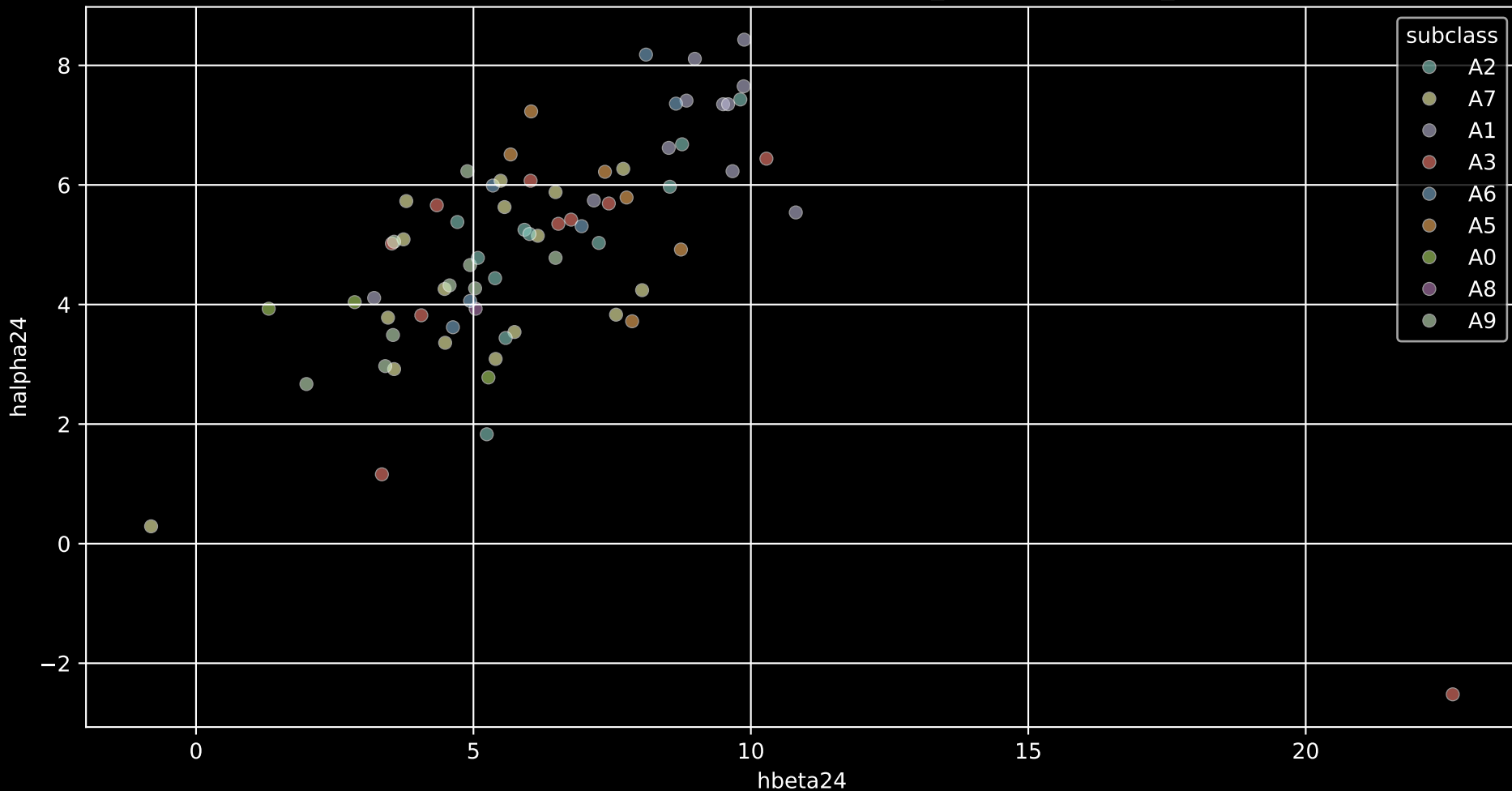
Star Field (planid) [GAC_080935N290534_M1]



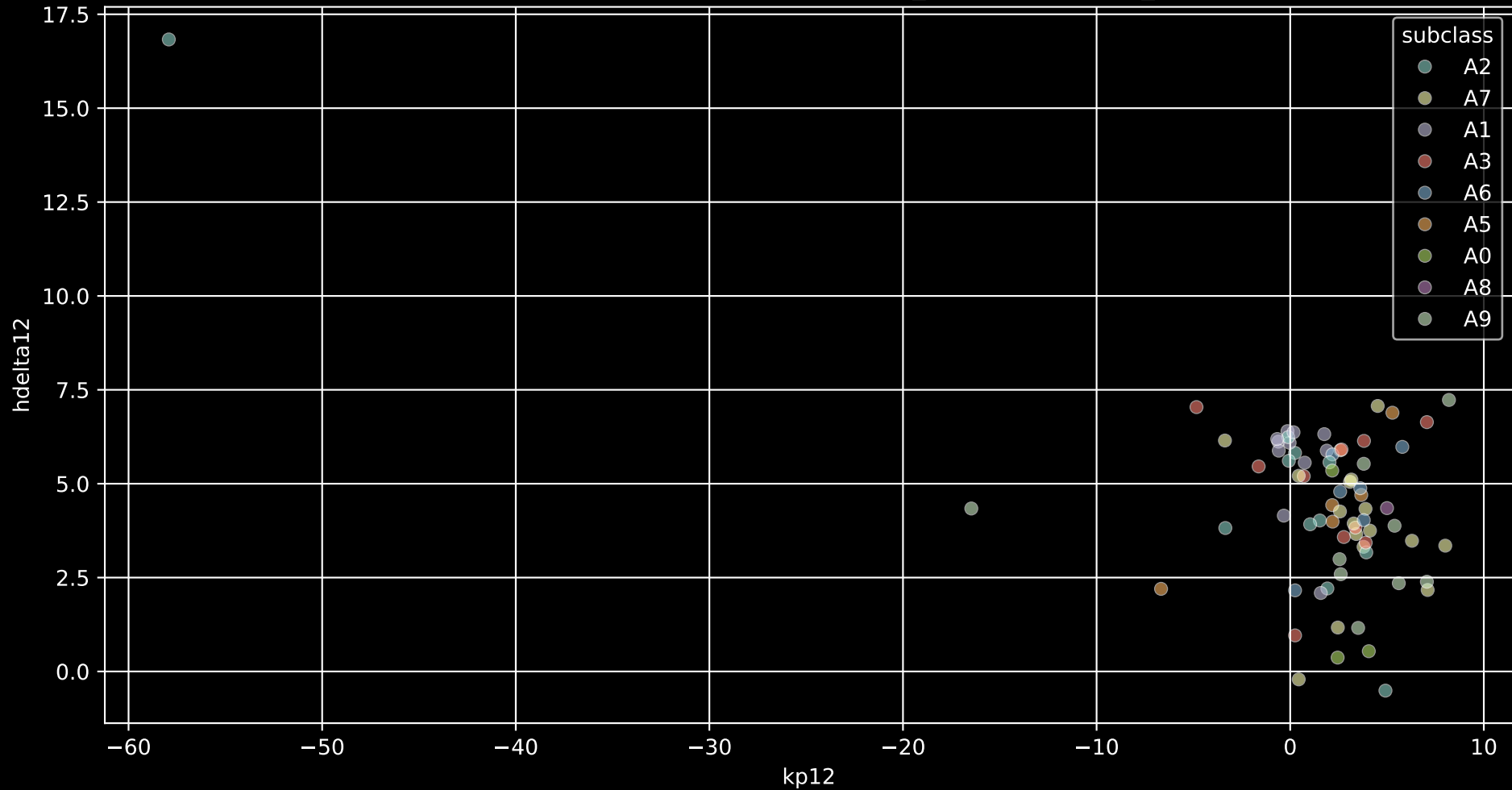
hdelta24 vs hgamma24 (H δ variant vs H γ variant; both Balmer diagnostics). (planid GAC_080935N290534_M1)



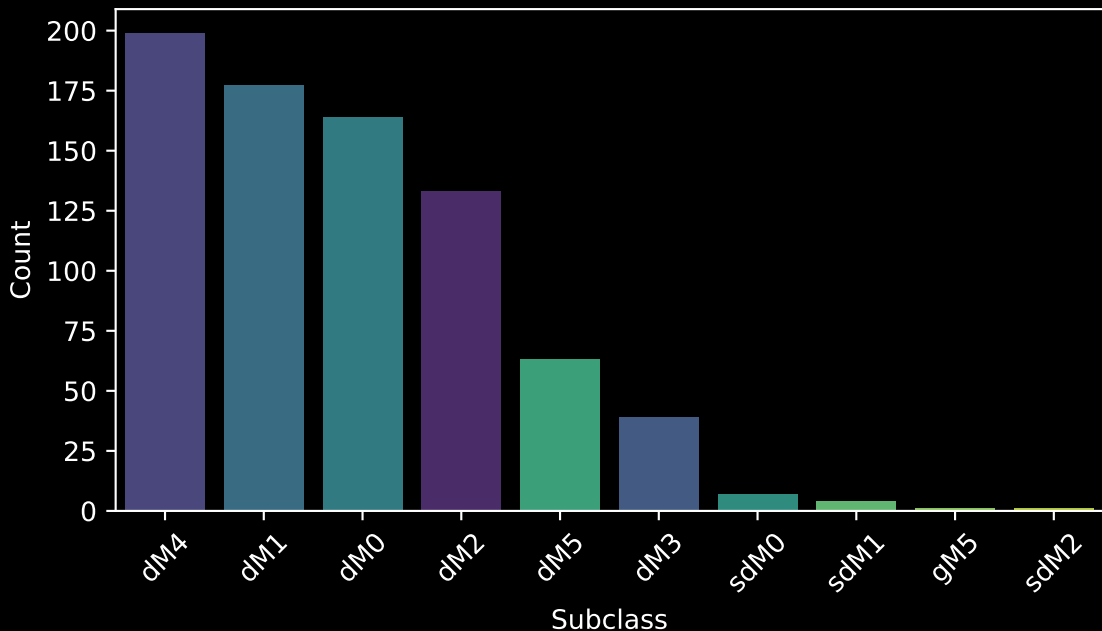
Expect a monotonic increase with cooling (A1 to A9),
but non-linear deviations highlight chromospheric activity or veiling (e.g., emission in H α for active stars)
A types: hbeta24 vs halpha24 (planid GAC_080935N290534_M1)



Metals vs Balmer would show anti-correlation in peculiar A stars (weaker Balmer in metal-rich).
For this field, it adds abundance insights, revealing if metal-poor halo interlopers (~ -3 [M/H] tail) affect A stars
A types: kp12 vs hdelta12 (planid GAC_080935N290534_M1)

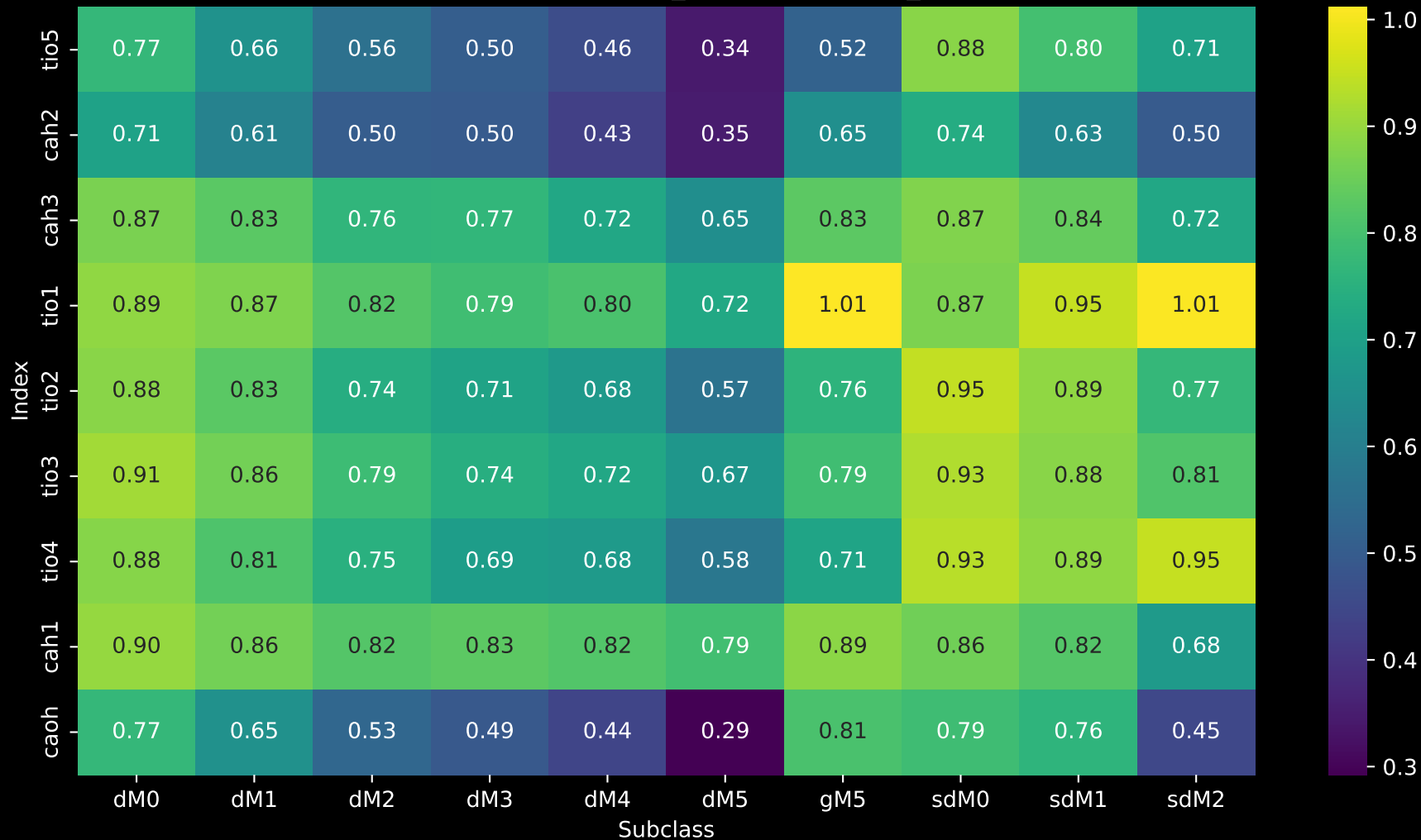


Subclass distribution for planid GAC_080935N290534_M1
(n= 788)

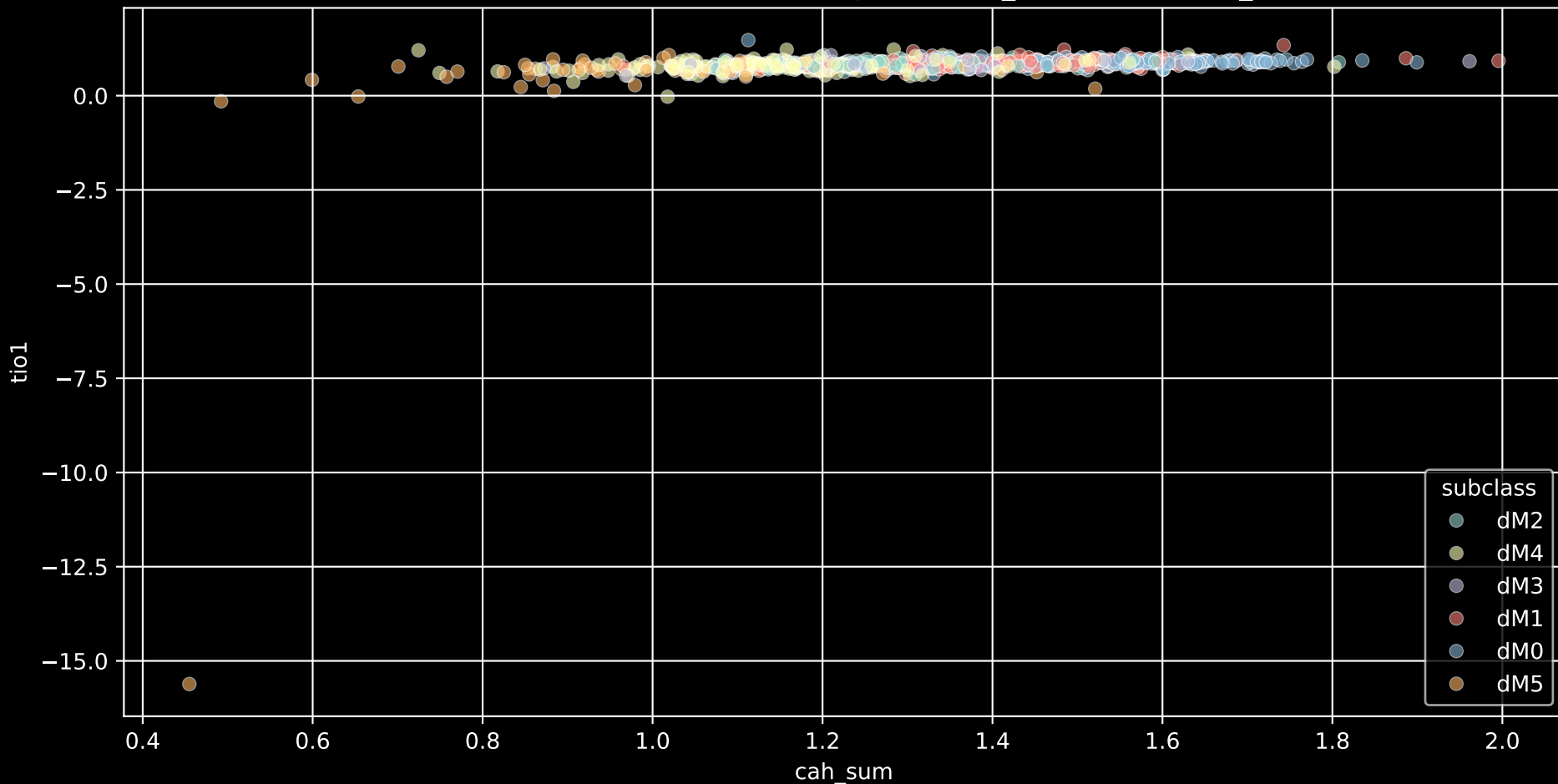


LAMOST DR11 Median Spectra Indices By Subclass

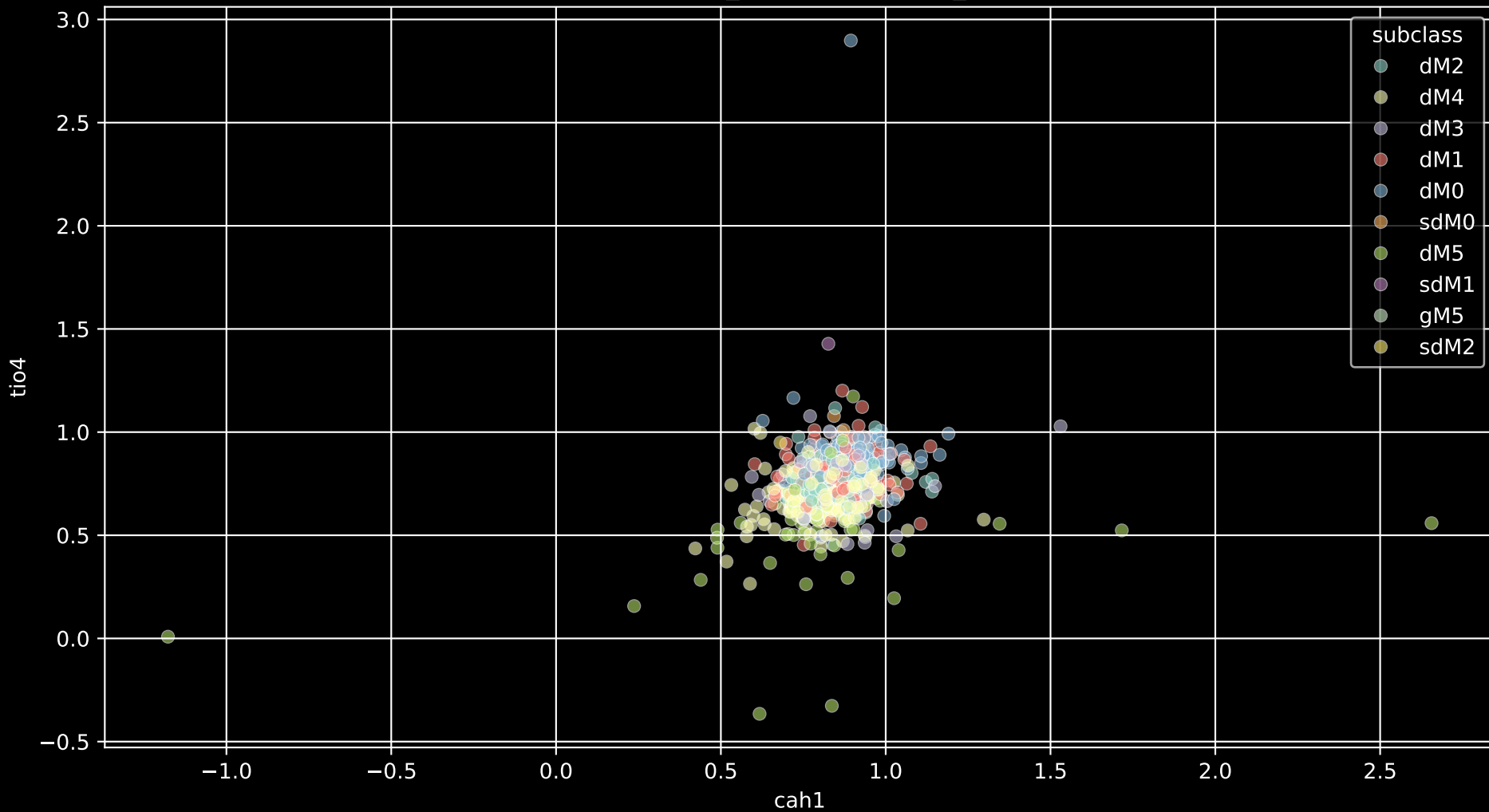
Star Field (planid) [GAC_080935N290534_M1]



This diagnostic diagram separates M dwarfs (strong CaH, moderate TiO) from giants (weaker CaH).
For this field, it would confirm all dM are dwarfs (no giant contamination in the ~ 7 Gyr median age population),
show anti-correlation with subtype (CaH weakens, TiO strengthens from dM0 to dM6)
(cah2 + cah3) vs tio1 (M dwarfs in planid GAC_080935N290534_M1)



cah1 vs tio4 (CaH variant vs another TiO; alternative for gravity/metallicity)
planid GAC_080935N290534_M1



cah1 vs tio2 (CaH variant vs TiO; for abundance vs temperature)
planid GAC_080935N290534_M1)

